



MASTER TRAINER SLIDES · NON-WSQ

Generative AI for Project Management

Course Code: C013

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COURSE ADMINISTRATION

Welcome & Housekeeping

About the Trainer



Your Trainer

General Trainer template —
to be completed by the trainer

NAME

TITLE / DESIGNATION

QUALIFICATIONS

AREAS OF EXPERTISE

TRAINING & INDUSTRY EXPERIENCE

CONTACT

About the Trainer



Dr Alfred Ang

Principal Trainer
Tertiary Infotech Academy Pte Ltd

ROLE

Principal Trainer, Tertiary Infotech Academy Pte Ltd

EXPERTISE

Project management, generative AI adoption and digital transformation.

DELIVERS

Courses on project management, business analysis and applied generative AI.

FOUNDER

Founder and lead instructor at Tertiary Infotech / Tertiary Courses.

Let's Know Each Other

1

Who you are

Your name, organisation and role.

2

What you run

The kind of projects you manage — size, industry and team.

3

What you use

Which GenAI tools you already use, and what for.

4

What you want

One project task you would most like GenAI to take off your hands.

Ground Rules

1

Set your mobile phone to silent mode.

2

Participate actively — no question is too small.

3

Mutual respect: agree to disagree.

4

One conversation at a time.

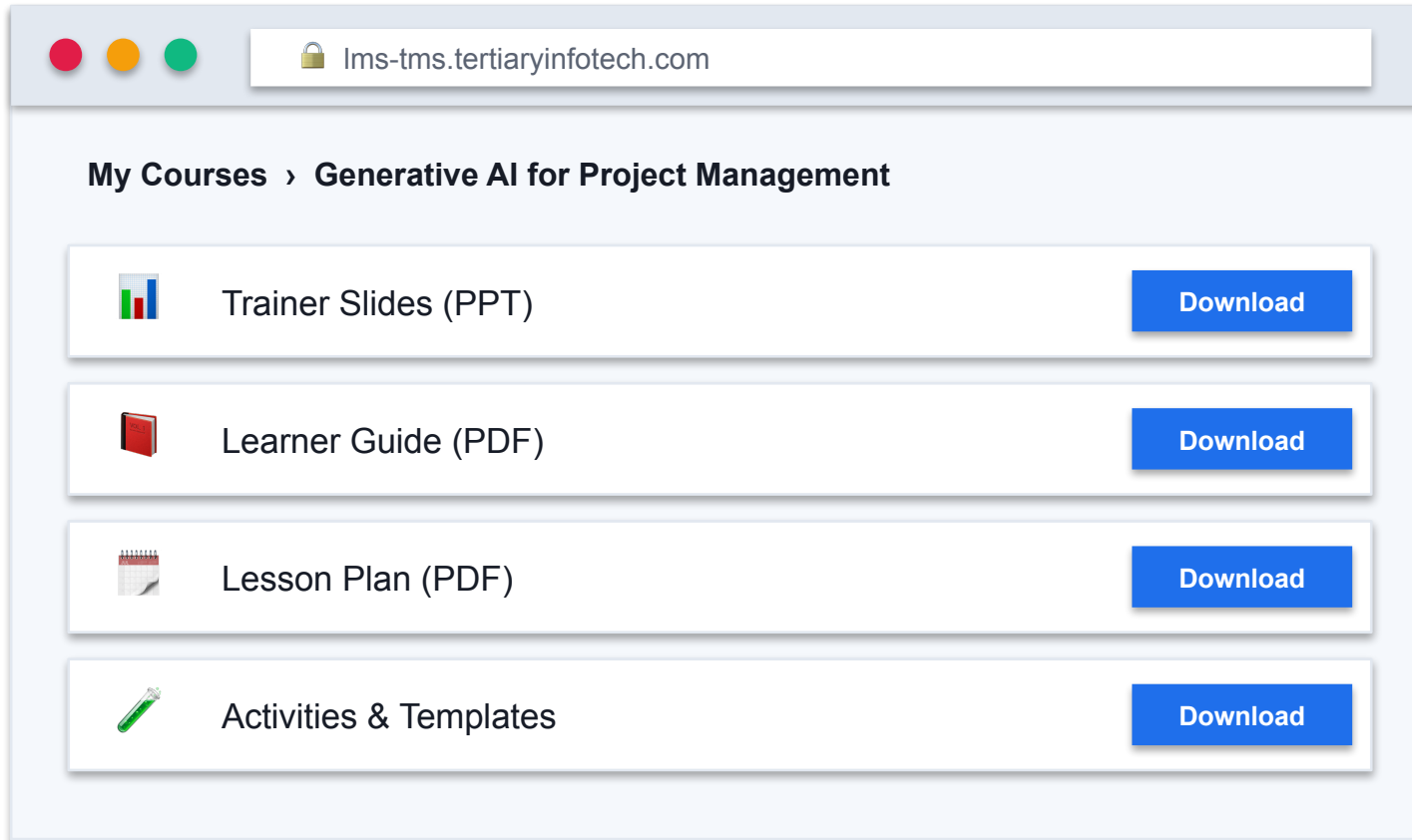
5

Be punctual; return from breaks on time.

6

Full attendance — attend both days to get the most from the course.

Download Your Course Material



- 1 Go to lms-tms.tertiaryinfotech.com
- 2 Log in with the account emailed to you
- 3 Open this course from My Courses
- 4 Download the slides, Learner Guide and templates
- 5 Keep them open — refer to them during the activities

Download the Project Management Templates

1

What's in the pack

Charter, RTM, WBS and WBS Dictionary, Gantt, schedule network diagram, RACI, SEAM, tornado chart, decision tree, check sheet and control chart.

2

Where to get it

The shared course template pack link is on the LMS and in your Learner Guide.

3

When to download

Before the first activity — you will use these templates throughout both days.

4

Formats

Excel, Word and PowerPoint — open them in Office or Google Workspace.

Course Outline

Topic 1 — Initiation & Planning

- Project & GenAI fundamentals
- 10 prompting patterns
- Charter & business case
- Stakeholders & RACI
- Requirements & RTM
- WBS & WBS dictionary
- Schedule & critical path
- Cost & budget

Topic 2 — Execution & Delivery

- Direct & manage work
- Manage project knowledge
- Quality & root cause analysis
- PDCA & DMAIC
- Optimise resources & budget
- Develop & manage the team
- Manage communications

Topic 3 — Risk & Closure

- Identify & analyse risk
- Tornado & decision trees
- Plan risk responses
- Procurement & contracts
- Monitor & control change
- Earned value analysis
- Control quality
- Close the project

Lesson Plan — 2 Days, 7.5 Instructional Hours per Day

Day 1

- Day 1 — Initiation & Planning with GenAI
 - Introduction, ground rules and course outline
 - Topic 1: Project Initiation and Planning with GenAI
 - Activities 1–10
 - Lunch 1:00–1:30pm
 - End of Day 1

Day 2

- Day 2 — Execution, Risk & Closure with GenAI
 - Day 1 recap
 - Topic 2: Project Execution and Delivery with GenAI (Activities 11–17)
 - Topic 3: Managing Project Risks with GenAI (Activities 18–25)
 - Course recap and feedback
 - End of Day 2



FOUNDATIONS

Projects, Project Management & GenAI

DEFINITION

What is a Project?

1

Creates something unique

A unique product, service or result — not business as usual.

2

Is time-limited

It has a definite beginning and a definite end.

3

Drives change

It moves the organisation from a current state to a future state.

4

Enables value creation

It delivers a business benefit that justifies the investment.

Examples of Projects

1

Developing a software application

2

Building a new office

3

Filming a motion picture

4

Creating an advertising campaign

5

Creating new processes within a business unit

6

Rolling out a GenAI assistant to a service team

Small Projects

What makes a project 'small'

- Typically less than six months
- Relatively limited resources
- Limited interdependency with other projects
- Limited strategic impact
- Relatively informal — reduced documentation and process
- The PM may have to lead AND work directly on the project
- May be scaled into a larger but separate project

Examples

- Develop a prototype to showcase an innovative idea
- Implement a jig for product qualification
- Construct a new pavement in the garden
- Set up a booth for an exhibition
- Pilot a GenAI tool with one team before rollout

REFLECTION

What small projects have you been involved in?

Turn to the person next to you. Share one project — what made it a project rather than routine work, and what made it hard.

DEFINITION

What is Project Management?

1

The practice

Initiating, planning, executing, controlling and closing the work of a team to achieve specific goals and meet specific success criteria at a specified time.

2

Why it exists

Projects power change and enable the organisation to achieve its business objectives.

3

Who is accountable

The project manager. GenAI changes HOW the work is done — never WHO is accountable for it.

CHANGE ONE, YOU CHANGE THEM ALL

The Triple Constraint

1

Scope

What the project will deliver.

2

Time

When it must be delivered.

3

Cost

What the organisation will spend.

4

Quality

Sits at the centre — change any one constraint and quality is what usually absorbs it.

Project Manager Competencies

Ways of Working

- Agile, Waterfall and hybrid
- Tailoring the method to the environment
- Knowing which approach fits

Power Skills

- Collaborative leadership
- Effective communication
- Influence without authority

Business Acumen

- Priority and cost vs benefit
- Making good business decisions
- Continuous learning

The 12 Project Management Principles

1

Be a diligent, respectful and caring steward

2

Create a collaborative project team environment

3

Effectively engage with stakeholders

4

Focus on value

5

Recognise, evaluate and respond to system interactions

6

Demonstrate leadership behaviours

7

Tailor based on context

8

Build quality into processes and deliverables

9

Navigate complexity

10

Optimise risk responses

11

Embrace adaptability and resilience

12

Enable change to achieve the envisioned future state

Project Success Depends On

1

Organisational project maturity

2

Project manager effectiveness

3

Funding and resource availability

4

Team member skill levels

5

Collaboration and communication with stakeholders

6

Understanding of the core problem and related needs

Project Life Cycles

How the project is structured

- Project Life Cycle
 - The phases the project passes through
 - May have phase gates — review performance against objectives
 - At each gate: proceed, change or stop

How you deliver

- Development Life Cycle
 - Predictive (Waterfall)
 - Iterative
 - Incremental
 - Agile
 - Hybrid

Waterfall vs Agile

Predictive

- Predictive / Waterfall
 - Requirements fixed up front
 - Performs each process once for the whole product
 - Change is controlled through change requests
 - Suits well-understood, stable work

Adaptive

- Adaptive / Agile
 - Requirements emerge and are re-prioritised
 - Performs each process once per increment
 - Change is expected and welcomed
 - Suits high-uncertainty work

Agile = Incremental + Iterative

1

Incremental

Small, regular releases of working value rather than one big-bang delivery.

2

Iterative

Reflecting and improving regularly — each cycle makes the product better.

3

High uncertainty

Agile suits work where the requirements cannot be fully known up front.

4

Many origins

Agile draws on Scrum, XP, Kanban, Lean and DSDM.

The Project Team

Makeup

- Team makeup varies
 - By organisational culture, location and scope
 - Full-time or part-time members
 - Includes generalists and specialists
 - Co-location helps — learning by osmosis

Skills

- Skill requirements
 - Ensure relevant skill sets are present
 - Avoid single points of failure
 - Create cross-functional teams
 - Use generalising specialists — 'T-shaped' people



GENERATIVE AI

GenAI for the Project Manager

What is Generative AI?

1

Creates new content

GenAI tools generate text, images, code, tables and ideas from a prompt.

2

Trained on patterns

They predict likely output from patterns in training data — they do not 'know' your project.

3

Conversational

You refine the output by refining the prompt — it is a dialogue, not a search box.

4

Confidently wrong

They will produce fluent, plausible output that is factually wrong. Verification is your job.

The GenAI Tools You'll Use

1

ChatGPT

chat.openai.com — general drafting, tables, analysis.

2

Claude

claude.ai — long documents, structured reasoning, critique.

3

Gemini

gemini.google.com — research-backed drafting and Workspace integration.

4

Microsoft Copilot

copilot.microsoft.com — drafting inside Office documents.

5

Project Manager Buddy GPT

A purpose-built PM GPT used throughout this course.

Project Manager Buddy GPT

1

What it is

A purpose-built GPT for project management, used throughout this course's activities.

2

Why use it

It already knows PM terminology, artifacts and process — so your prompts can be shorter.

3

Where to find it

Open it from the link in your Learner Guide, or search 'Project Manager Buddy' in the GPT store.

4

Still verify

It is still a GenAI tool: it will invent figures and sources. Check everything.

WHY GenAI FOR PROJECT MANAGEMENT



The goal of GenAI is to take a project professional's mundane tasks and make them easy, like 'here you go, this is done.' Learning to leverage it ultimately increases the value of the project professional to the organization, because not only is it saving the professional's precious time, but it is also empowering the professional to focus on strategic aspects of the job that no one else can do.

John Signo, VP DMT Delivery Lead, Bank of America, USA

WHY GenAI FOR PROJECT MANAGEMENT



I don't think GenAI can run a project, but as a reviewer of project work, it can find errors much faster than humans — reviewing schedules, plans and project boards to identify risks, scheduling challenges, unknown dependencies and mistakes.

John Signo, VP DMT Delivery Lead, Bank of America, USA

WHY GenAI FOR PROJECT MANAGEMENT



I use GenAI to augment complex decision-making — to see things from multiple angles, find innovative approaches and get intelligent suggestions I would not have reached alone.

Yuliya Zhevno, Project and Change Management Consultant, Re:Work Consulting, Kazakhstan

Where GenAI Helps a Project Manager Most

1

Drafting

Charters, plans, reports, communications — a good first draft in seconds.

2

Structuring

Turning messy input into a WBS, RTM, risk register or RACI matrix.

3

Ideation

Generating risks, requirements, options and root causes you would not think of.

4

Reviewing

Auditing your own plans for gaps, ambiguity and missing dependencies.

5

Analysing

Comparing alternatives and explaining the trade-offs in a table.

6

Communicating

Rewriting the same message for very different audiences.

What GenAI Does Well — and Badly

Use it for this

- Does well
 - Producing a structured first draft fast
 - Generating breadth — many options and ideas
 - Reformatting and summarising
 - Explaining a concept or technique
 - Reviewing your work for gaps

Never trust this

- Does badly
 - Anything requiring facts about YOUR organisation
 - Arithmetic — verify every calculation
 - Citing real sources — it invents them
 - Judging what matters in your context
 - Anything about a specific named person

THE GOLDEN RULE

You remain accountable for every output.

GenAI drafts; you decide. Every figure, source and recommendation that leaves your hands is yours — not the model's.

Using GenAI Responsibly at Work

1

Confidentiality

Never paste client data, personal data or commercially sensitive information into a public tool.

2

Check your policy

Your organisation may mandate an enterprise tool with data protections.

3

Verify facts

Treat every figure, date and citation as unverified until you check it.

4

Disclose appropriately

Be transparent where AI materially shaped a deliverable.

5

Bias

Output reflects patterns in training data — review recommendations about people especially carefully.

6

Keep the human in the loop

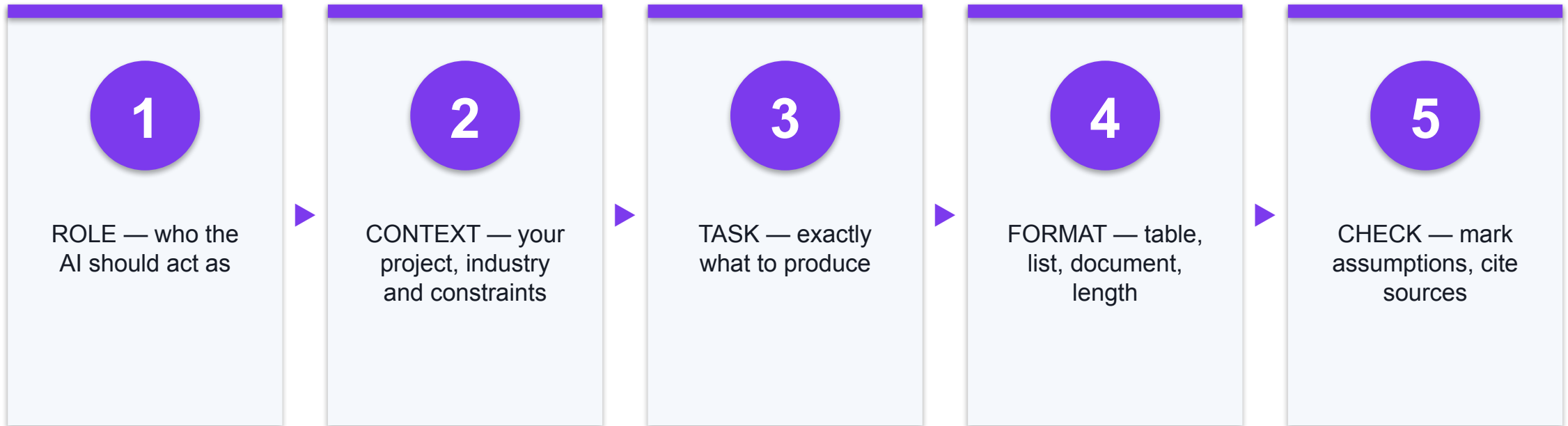
Approval decisions stay with people, not models.



PROMPTING

Ten Prompting Patterns for Project Managers

Anatomy of a Good Prompt



The Ten Patterns

1

Persona

Frame the output from the perspective of a specific project role — or a non-human entity such as project management software.

3

Flipped Interaction

Let GenAI interview you — it asks the questions, one at a time, to find the gaps in your thinking.

5

Cognitive Verifier

Force GenAI to break a complex question into smaller sub-questions and then combine the answers.

7

Visualisation Generator

Have GenAI write the prompt for a text-to-image or diagram tool so you can visualise project information.

9

Risk Matrix Template

A long-form template that produces a complete, industry-specific risk matrix in one pass.

2

Question Refinement

Ask GenAI to improve your question when there is no clarity about the problem you are trying to solve.

4

Alternative Approaches

Generate several options, then compare their pros and cons before you decide.

6

Fact Check List

For outputs that rest on external information, require the facts to be listed separately with links to sources.

8

Reflection

Ask for the reasoning and assumptions behind a recommendation — essential for risk and people decisions.

10

Decision Making Template

Decompose a complex decision into smaller questions that together answer it.

PROMPTING PATTERN

Pattern — Persona

Frame the output from the perspective of a specific project role — or a non-human entity such as project management software.

EXAMPLE PROMPT

Act as a stakeholder of the project. Perform an assessment of the project charter and provide a list of recommendations to improve scope clarity and completeness.

PROMPTING PATTERN

Pattern — Question Refinement

Ask GenAI to improve your question when there is no clarity about the problem you are trying to solve.

EXAMPLE PROMPT

Within the scope of creating a cost-benefit analysis of outsourcing (total cost: US\$60,000) versus in-house hiring (total cost: US\$70,000), suggest a better version of the question to be used.

PROMPTING PATTERN

Pattern — Flipped Interaction

Let GenAI interview you — it asks the questions, one at a time, to find the gaps in your thinking.

EXAMPLE PROMPT

I would like you to ask questions to help with my preparation for a project update with my stakeholders. Ask me questions one at a time so you can build the next question based on my responses.

Pattern — Alternative Approaches

Generate several options, then compare their pros and cons before you decide.

EXAMPLE PROMPT

I am a construction project manager experiencing a delay in completion of some tasks. I want to identify the root cause(s) of the problem. Suggest at least two options to deal with the problem(s) identified and compare the pros and cons of each. Present the results in a table.

PROMPTING PATTERN

Pattern — Cognitive Verifier

Force GenAI to break a complex question into smaller sub-questions and then combine the answers.

EXAMPLE PROMPT

A software development project has a US\$100,000 budget and a timeline of six months. After three months the actual cost is US\$60,000 and the project is 50% complete. What are the risks of not completing on time and on budget? Am I over or under budget? Am I behind or ahead of schedule? What is the earned value and estimate to complete? Combine the answers to produce the final answer to the first question.

Pattern — Fact Check List

For outputs that rest on external information, require the facts to be listed separately with links to sources.

EXAMPLE PROMPT

Generate a list of the 10 most common frequent-flier challenges in the airline industry based on facts. The set of facts should be inserted at a specific point in the output and correctly show links to sources.

Pattern — Visualisation Generator

Have GenAI write the prompt for a text-to-image or diagram tool so you can visualise project information.

EXAMPLE PROMPT

I want to generate a prompt that I can provide to a diagram tool to visualise the 10 most common sources of project scope creep for my project kickoff presentation.

PROMPTING PATTERN

Pattern — Reflection

Ask for the reasoning and assumptions behind a recommendation — essential for risk and people decisions.

EXAMPLE PROMPT

I need to provide feedback to a project team member who made a mistake with financial implications. Explain the reasoning and assumptions behind your recommendation on how to approach this meeting with them.

Pattern — Risk Matrix Template

A long-form template that produces a complete, industry-specific risk matrix in one pass.

EXAMPLE PROMPT

Imagine you are an expert project manager in the [Industry], focusing on [Project type] projects. Develop a comprehensive risk matrix. For each risk include: Risk name, Description, Level of impact (high/medium/low) and Likelihood (high/medium/low). Cover budget, schedule, technology, regulatory, resource, scope creep, stakeholder, data security, quality and external dependency categories. Output as a table.

Pattern — Decision Making Template

Decompose a complex decision into smaller questions that together answer it.

EXAMPLE PROMPT

Imagine you are an expert project manager in the [Industry] working on a [Project type] project. Break the complex question "[Complex question?]" into smaller questions that collectively answer it. Consider the range of outcomes, sources of resistance, risks and mitigation plans, and key milestones. Provide one-to-three-sentence answers to each, then ask me where I would like to go deeper.

Prompting Patterns for Project Managers

ACTIVITY 1

Work through the ten prompting patterns on a project of your own. You will see the same question answered badly by a vague prompt and well by a structured one, and you will keep a personal prompt library for the rest of the course.

You'll produce

A personal prompt library of 10 tested prompts, each with the output it produced and your critique.

Tools: ChatGPT · Claude · Gemini · Copilot · Project Manager Buddy GPT

Step-by-step instructions: [see your Learner Guide](#)

Prompting Patterns for Project Managers

ACTIVITY 1

PROMPT GenAI WITH

Act as a stakeholder of the project. Perform an assessment of the project charter and provide a list of recommendations to improve scope clarity and completeness.

⚠ THEN CHECK IT

GenAI gives you a fast first draft. Check every figure, source and assumption before you use it.

Prompting Patterns for Project Managers

Check your work

You have 10 prompts in your library, each with output and a critique, and you can state one concrete case where GenAI produced a plausible but wrong answer that you caught.



COURSE ED-TOOLS

Five Collaborative Project Management Tools

BROWSER-BASED · REAL-TIME · NO INSTALL

The Ed-Tools You'll Use

1

RACI Collaborative Matrix

Drag A / R / C markers into a live responsibility matrix with your team.

2

Business Model Canvas Studio

The nine-block Osterwalder canvas — validate the business model behind your project.

3

Design Thinking Studio

Empathise, Define, Ideate, Prototype, Test — with dot-voting on ideas.

4

Scrum Agile Transformation Studio

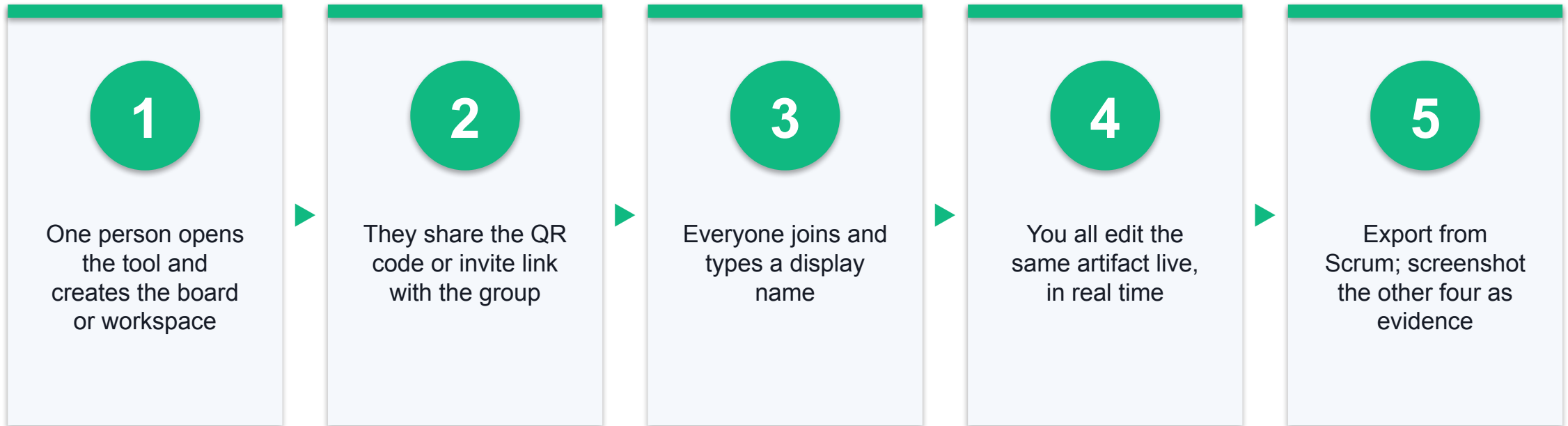
Sprint planning, planning poker, burndown and retrospectives; exports JSON/CSV.

5

Digital AI Transformation Roadmap Studio

Initiatives, impact-vs-effort matrix, 12-month timeline, risk register and KPIs.

How the Ed-Tools Work



RACI Collaborative Matrix

Build a responsibility assignment matrix with your team by dragging role markers into a live grid.

<https://alfredang.github.io/raci/>

Drag A / R / C

Drag markers into cells; click a marker to remove it, or drag to the trash.

Add roles & activities

'+ Add role' and '+ Add activity'; click any header to rename it.

Pre-seeded

A GenAI project example with 4 roles and 5 activities is already loaded.

Share

QR code and board URL — everyone edits the same matrix live.

Business Model Canvas Studio

Design and validate the business model behind your project on the nine-block Osterwalder canvas.

<https://alfredang.github.io/bcm/>

Nine blocks

Partners, Activities,
Resources, Value
Propositions, Relationships,
Channels, Segments, Costs,
Revenue.

Sticky notes

Title, description, category,
priority — added to any
block.

Validate

Set each note Draft →
Validate → Confirmed as
evidence firms up.

Guidance

The (i) button on each block
reveals coaching questions.

Design Thinking Studio

Run the five design thinking stages to elicit and prioritise requirements as a group.

<https://alfredang.github.io/designthinking/>

Five stages

Empathize → Define →
Ideate → Prototype → Test.

Empathy map

Says / Thinks / Does / Feels
quadrants for your persona.

Dot voting

Voting is enabled on the
ideation and prioritisation
sections.

Notes

Title, description and
category; some sections
support resolve.

Scrum Agile Transformation Studio

Plan sprints, run ceremonies, estimate with planning poker and hold retrospectives together.

<https://alfredang.github.io/scrum/>

Pick a role

Product Owner, Scrum Master, Developer, Stakeholder or Agile Coach.

Planning poker

Reveal, reset and commit estimates as a team.

Ceremonies

Sprint planning, burndown tracking and retrospectives.

Exports

JSON, CSV (stories + tasks) or a printable HTML report.

Digital AI Transformation Roadmap Studio

Build a transformation roadmap with initiatives, prioritisation, risks and KPIs.

<https://alfredang.github.io/digitaltransformation/>

Seven tabs

Overview, Phases & Initiatives, Priority Matrix, Maturity, Timeline, Risks, KPIs.

Impact vs Effort

Quick Wins, Major Projects, Fill-ins and Thankless quadrants.

Risk register

Likelihood, impact, computed score, mitigation and owner.

KPIs & timeline

Current vs target values on a 12-month timeline.

TOPIC 01

Project Initiation and Planning with GenAI

Charter · business case · stakeholders · scope · WBS · schedule · cost



Key Concepts — Topic 01

1

What a project is

A unique product, service or result; time-limited; drives change and enables value creation.

2

GenAI as your planning co-pilot

ChatGPT, Claude, Gemini and Copilot draft charters, WBS, schedules and estimates in minutes — you stay accountable for the judgement.

3

Ten prompting patterns

Persona, Question Refinement, Flipped Interaction, Alternative Approaches, Cognitive Verifier, Fact Check List, Visualisation Generator, Reflection and the risk/decision templates.

4

Project Charter

Authorises the project, names the sponsor and PM, and fixes objectives, budget and success criteria.

5

Business case & value

Cost-benefit analysis and a Business Model Canvas prove economic feasibility before work starts.

6

Scope & requirements

Collect requirements, build the Requirements Traceability Matrix, write the scope statement, decompose into a WBS.

7

Schedule

Define, sequence and estimate activities; build the network diagram and find the critical path.

8

Cost

Analogous, parametric, bottom-up and three-point estimating; the cost baseline plus contingency and management reserves.

GenAI Activities — Topic 01

Activities 1–4

- Prompting Patterns for Project Managers
- Business Case and Cost-Benefit Analysis with GenAI
- Develop the Project Charter with GenAI
- Identify and Map Stakeholders with GenAI

Activities 5–8

- Assign Responsibilities with the RACI Ed-Tool
- Collect Requirements with GenAI and Design Thinking
- Build the Requirements Traceability Matrix with GenAI
- Decompose Scope into a WBS and WBS Dictionary with GenAI

Activities 9–10

- Build the Schedule and Find the Critical Path with GenAI
- Estimate Costs and Build the Budget with GenAI



INITIATION

Develop the Project Charter

Project Integration Management

1

Business case first

A well-researched business case proves the project is worth doing.

2

Then the charter

An approved Project Charter formally authorises the project.

3

Then the plan

A well-planned Project Management Plan integrates scope, schedule, cost and quality.

4

Key benefit

Deliver business value aligned with the organisation's strategic objectives.

Business Value — Why Projects Are Approved

1

Financial gain

2

New customers

3

Social benefit

4

First to market

5

Improvement — technological or process

6

Regularisation — compliance with standards and regulations

The Business Case

- A documented economic feasibility study that establishes the benefits of the project work.
- It provides the basis for authorisation of further project activities.
- Cost-benefit analysis quantifies the costs against the expected benefits over time.
- The Business Model Canvas tests whether the value proposition and the economics hold together.

Business Case and Cost-Benefit Analysis with GenAI

ACTIVITY 2

Before a project is authorised somebody has to prove it is worth doing. You will use GenAI to draft a cost-benefit analysis and a one-page business case, then validate the business model behind it on the Business Model Canvas Studio.

You'll produce

A completed cost-benefit analysis, a one-page business case, and a validated 9-block Business Model Canvas.

Tools: GenAI tool · Business Model Canvas Studio (alfredang.github.io/bcm) · Cost Benefit Analysis.xlsx · Business Case on a Page.docx

Step-by-step instructions: see your [Learner Guide](#)

Business Case and Cost-Benefit Analysis with GenAI

ACTIVITY 2

PROMPT GenAI WITH

List the one-off and recurring costs, and the quantified annual benefits, of [your project] over 3 years. Present the result as a table with a column explaining how each figure will be verified.

⚠ THEN CHECK IT

GenAI gives you a fast first draft. Check every figure, source and assumption before you use it.

Business Case and Cost-Benefit Analysis with GenAI

Check your work

Your canvas has all nine blocks populated, every note carries a Draft/Validate/Confirmed status, and your business case states a payback period you can defend with your own figures.

WHAT IT DOES

The Project Charter

1

Authorises the project

It is what allows the project to formally begin.

2

Empowers the PM

It gives the project manager the right to apply organisational resources.

3

Defines the rationale

It states the business need the project answers.

4

Verifies alignment

It confirms the project supports the organisation's strategic goals.

5

Creates focus

It keeps everyone aligned to a single, clear project vision.

6

Is signed

An unsigned charter is not a charter.

What's Included in a Project Charter

1

Names

Project sponsor, project manager and key stakeholders.

2

Project description

Preliminary requirements and measurable objectives.

3

Business need

Including financial goals and milestones.

4

Summary schedule

High-level milestones.

5

Assumptions & constraints

Boundaries, overall risk, approval requirements, approved budget.

6

From the business case

Success criteria and exit criteria.

Project Vision Statement

- Created by the project sponsor or an executive.
- Includes a clear vision of the desired objectives and how they align with the organisation's strategic goals.
- Refer back to it throughout the project to maintain alignment when pressure to change mounts.

Develop the Project Charter with GenAI

ACTIVITY 3

The charter authorises the project and gives the PM the right to apply resources. You will generate a full charter with GenAI and then correct it against what a charter must actually contain.

You'll produce

An approved-quality Project Charter for your group's project.

Tools: GenAI tool · Project Manager Buddy GPT · Project Charter Template.xlsx

Step-by-step instructions: [see your Learner Guide](#)

Develop the Project Charter with GenAI

ACTIVITY 3

PROMPT GenAI WITH

Write a project charter for launching an internal knowledge-sharing platform. Include objectives, scope, risks, stakeholders and success criteria. Mark every assumption clearly.

⚠️ THEN CHECK IT

GenAI gives you a fast first draft. Check every figure, source and assumption before you use it.

Develop the Project Charter with GenAI

Check your work

Your charter names a sponsor and PM, has measurable objectives and explicit success AND exit criteria, and contains no unmarked AI-invented figures.

The Kickoff Meeting

Purpose

- Establish project context
- Assist in team formation
- Align team and stakeholders with the vision

Organisational / Public

- Announce project initiation
- Share the high-level vision, purpose and value
- Identify sponsor, key stakeholders and PM
- Cover high-level charter items

Internal / Team

- Give a charter overview
- Clarify roles and responsibilities
- Present planning results
- Initiate the product backlog



PLANNING

The Project Management Plan

Planning Starts with the Project Management Plan

1

Enables the PM

To execute, monitor, control and close the project.

2

Establishes guardrails

So controls are maintained as the project runs.

3

Allows tailoring

Teams can adapt their way of working and act quickly and flexibly.

4

Integrates everything

Scope, schedule, cost, quality, resource, communications, risk, procurement and stakeholders.

What You Plan

1

Scope — what is and is not included

2

Schedule — when the work happens

3

Resources — who and what is needed

4

Budget — what it will cost

5

Risks — what could go wrong or right

6

Quality — how good it must be

Collaborative and Rolling Wave Planning

Plan with the team

- Collaborative planning
 - The product owner decides objectives per customer needs
 - The team executes the work and helps plan it
 - Team members are the local domain experts
 - The PM / scrum master helps the team focus

Plan progressively

- Rolling wave planning
 - A form of progressive elaboration
 - Plan near-term work in detail, later work at a high level
 - Applied to work packages, planning packages and releases
 - Used in both adaptive and predictive approaches



STAKEHOLDERS

Identify and Engage Stakeholders

DEFINITION

Who is a Stakeholder?

- A stakeholder is anyone affected by your project — or who believes they are affected.
- The outcome you want: productive relationships that ensure stakeholders support the project objectives.
- The process: Identify → Analyse → Prioritise → Engage → Monitor.

Project Stakeholders

1

End users

2

Customers

3

Employees

4

The organisation itself

5

Managers

6

Sponsors

7

Business partners

8

Suppliers and contractors

9

Government and regulators

10

The community

The Stakeholder Register

1

Name

Who they are.

2

Role

Their role on or around the project.

3

Power / Interest / Influence

How much they can affect the project, and how much they care.

4

Expectations and needs

What they want from the project.

5

Communication requirements

What they need to know, how and how often.

6

Engagement level

Unaware, Resistant, Neutral, Supportive or Leading.

Analysing Stakeholders

1

2D grids

Power/Interest, Power/Influence, Impact/Influence.

2

3D grid

The stakeholder 'cube'.

3

Salience model

Power, Legitimacy and Urgency combined.

4

Directions of influence

Upward, Downward, Sideward, Outward.

Directions of Influence

1

Upward

Senior management, the sponsor, the steering committee.

2

Downward

The project team and specialists providing knowledge and skills.

3

Sideward

Peers of the PM — other managers competing for the same resources.

4

Outward

Suppliers, users, government departments and the public.

Identify and Map Stakeholders with GenAI

ACTIVITY 4

Projects fail on stakeholders more often than on technology. You will use GenAI to generate a stakeholder register, then map power against interest and decide an engagement strategy for each.

You'll produce

A stakeholder register, a power/interest map and a Stakeholder Engagement Assessment Matrix.

Tools: GenAI tool · Stakeholder Impact Over Influence Matrix.xlsx · Salience Model Template.pptx · Stakeholder Engagement Assessment Matrix.xlsx

Step-by-step instructions: [see your Learner Guide](#)

Identify and Map Stakeholders with GenAI

ACTIVITY 4

PROMPT GenAI WITH

Identify potential stakeholders and their influence and interest levels for a new city-wide bike-sharing system. Explain the evidence needed to validate each rating.

⚠ THEN CHECK IT

GenAI gives you a fast first draft. Check every figure, source and assumption before you use it.

Identify and Map Stakeholders with GenAI

Check your work

Every stakeholder has a quadrant, a salience category, a current and desired engagement level, and one named action to close the gap.

MARK C FOR CURRENT, D FOR DESIRED

Stakeholder Engagement Assessment Matrix (SEAM)

1

Unaware

Does not know the project exists or its potential impact.

2

Resistant

Aware, and opposed to the change.

3

Neutral

Aware, but neither supportive nor resistant.

4

Supportive

Aware and supportive of the change.

5

Leading

Actively engaged in making the project succeed.

Allocating Responsibilities

1

Responsibility Assignment Matrix

Shows which team members are responsible for which work packages — most commonly a RACI.

2

Team Charter

Created with the team; establishes values, agreements and operating guidelines.

3

Resource Breakdown Structure

An organisational breakdown of the team showing where everyone fits.

4

Resource Calendars

Show when each resource is actually available, including leave and holidays.

Assign Responsibilities with the RACI Ed-Tool

ACTIVITY 5

A RACI matrix ends the 'I thought you were doing it' conversation. You will build one collaboratively in a live browser tool, using GenAI to propose the first pass.

You'll produce

A completed RACI matrix for your project, agreed by the whole group.

Tools: RACI Collaborative Matrix (alfredang.github.io/raci) · GenAI tool

Step-by-step instructions: see your [Learner Guide](#)

Assign Responsibilities with the RACI Ed-Tool

ACTIVITY 5

PROMPT GenAI WITH

For a project with these activities [paste] and these roles [paste], propose a RACI assignment. Give exactly one Accountable role per activity and explain any ambiguous accountability.

⚠ THEN CHECK IT

GenAI gives you a fast first draft. Check every figure, source and assumption before you use it.

Assign Responsibilities with the RACI Ed-Tool

Check your work

Every activity row has exactly one A, at least one R, and your group can justify each assignment; you have a screenshot saved as evidence.



SCOPE

Collect Requirements & Define Scope

The Scope Management Plan

1

Prepares the scope statement

Defines the process for producing a detailed project scope statement.

2

Enables the WBS

Establishes how the WBS is created from the scope statement.

3

Maintains the baseline

States how the scope baseline is approved and maintained.

4

Governs acceptance

Specifies how formal acceptance of deliverables is obtained.

Scope Planning Techniques

1

Document analysis

Derive requirements from existing documents.

2

Alternatives analysis

Identify options and approaches for executing the work.

3

Product analysis

Ask questions about the product — its use and characteristics.

4

Expert judgment

Draw on those who have done this work before.

Collect Requirements — Techniques

1

Data gathering

Brainstorming, interviews, focus groups, questionnaires, surveys, benchmarking.

2

Interpersonal & team skills

Nominal group technique, observation, facilitation.

3

Data analysis

Document analysis, alternatives analysis, product analysis.

4

Decision making

Voting, multi-criteria decision analysis.

5

Data representation

Mind mapping, affinity diagram, context diagram, use case diagram.

6

Expert judgment

Subject matter experts who know the domain.

Ideation & Structuring Techniques

1

Brainstorming

Combines a relaxed, informal approach with lateral thinking to generate many ideas.

2

Affinity diagram

Classifies large numbers of ideas into themed groups for review and analysis.

3

Nominal group technique

Individuals rank shortlisted ideas privately, then scores are tallied.

4

Mind mapping

Consolidates ideas into a single map showing commonality and difference.

5

Context diagram

Shows the system at the centre and its interactions with external entities.

6

Benchmarking

Compares your practices against others to identify and adopt best practice.

Collect Requirements with GenAI and Design Thinking

ACTIVITY 6

Requirements gathering is where GenAI earns its keep. You will run brainstorming, affinity clustering and prioritisation — using the Design Thinking Studio to structure the work and dot-vote as a group.

You'll produce

A clustered, prioritised requirements set with a MoSCoW classification.

Tools: Design Thinking Studio (alfredang.github.io/designthinking) · GenAI tool · MoSCoW with Sliders Template.xlsx

Step-by-step instructions: see your Learner Guide

Collect Requirements with GenAI and Design Thinking

ACTIVITY 6

PROMPT GenAI WITH

Create three user personas and their top pain points for [your project's users]. Then list 15 to 20 ideas for solving the highest-priority pain point. Mark assumptions that require user research.

⚠️ THEN CHECK IT

GenAI gives you a fast first draft. Check every figure, source and assumption before you use it.

Collect Requirements with GenAI and Design Thinking

Check your work

Your workspace has personas, an empathy map, a problem statement, a clustered idea board with vote counts, and a MoSCoW list where you changed at least one AI classification.

Prioritisation Techniques

Technique	How it works	Best for
MoSCoW	Must have / Should have / Could have / Won't have (for now)	Timeboxed agile delivery; comparing several points of view
Kano Model	Delighters, Satisfiers, Dissatisfiers, Indifferent	Prioritising by what most influences customer satisfaction
Paired Comparison	Rate and rank alternatives by comparing each against the other	A small range of subjective requirements
100 Points Method	Stakeholders distribute 100 points across requirements	Any size group; gives priority to stakeholder decision-making

The Requirements Traceability Matrix

1

Traces forward

From each requirement to the deliverable that satisfies it.

2

Traces backward

From each deliverable to the stakeholder need that justified it.

3

Links to tests

Every requirement has an acceptance criterion and a test case.

4

Proves completeness

At closure it proves every approved requirement was delivered.

Build the Requirements Traceability Matrix with GenAI

ACTIVITY 7

An RTM proves that what you built is what was asked for. GenAI can populate one quickly, but it will happily invent requirement sources — which is exactly the failure mode an RTM exists to prevent.

You'll produce

A complete Requirements Traceability Matrix with verified sources.

Tools: GenAI tool · Requirements Traceability Matrix.xlsx

Step-by-step instructions: [see your Learner Guide](#)

Build the Requirements Traceability Matrix with GenAI

ACTIVITY 7

PROMPT GenAI WITH

Generate 5 to 10 sample requirements for a smart-city parking app. Include Requirement ID, Description, Source, Acceptance Criterion and Test Case ID. Flag any requirement that is not testable.

⚠️ THEN CHECK IT

GenAI gives you a fast first draft. Check every figure, source and assumption before you use it.

Build the Requirements Traceability Matrix with GenAI

Check your work

Every row traces from a real, named source through to an acceptance criterion and a test case, with no orphan requirements or orphan tests.

WHAT IT CONTAINS

The Project Scope Statement

1

Scope description — of the project and the product

2

Acceptance criteria

3

The required deliverables

4

Out-of-scope items, stated explicitly for clarity

5

Constraints

6

Assumptions

Baselines

1

Scope baseline

The approved scope statement + WBS + WBS dictionary.

2

Schedule baseline

The approved version of the schedule.

3

Cost baseline

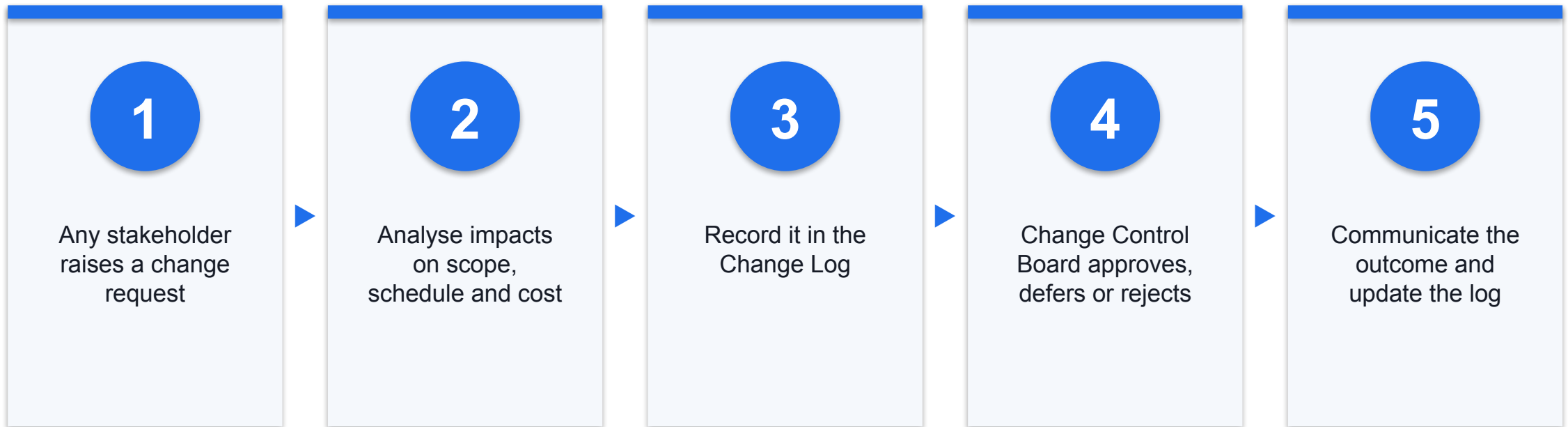
The approved time-phased budget.

4

Performance measurement baseline

The integrated scope, schedule and cost baseline you measure performance against.

The Change Request Process



Two Ways Scope Goes Wrong

Scope Creep

- Scope Creep
 - Scope or features are added that were not planned
 - Not approved through change control
 - Impacts cost and schedule
 - Often arrives as many small 'while you're at it' requests

Gold Plating

- Gold Plating
 - Delivering extra features the customer never asked for
 - Usually well-intentioned — the team adds 'value'
 - Consumes budget and schedule for no agreed benefit
 - Creates support and quality burden later

Decomposing Work into the WBS

1

Decomposition

Break the work down with your team from deliverables into the smallest useful unit.

2

Work package

A piece of work small enough to be assigned to one person or team.

3

WBS dictionary

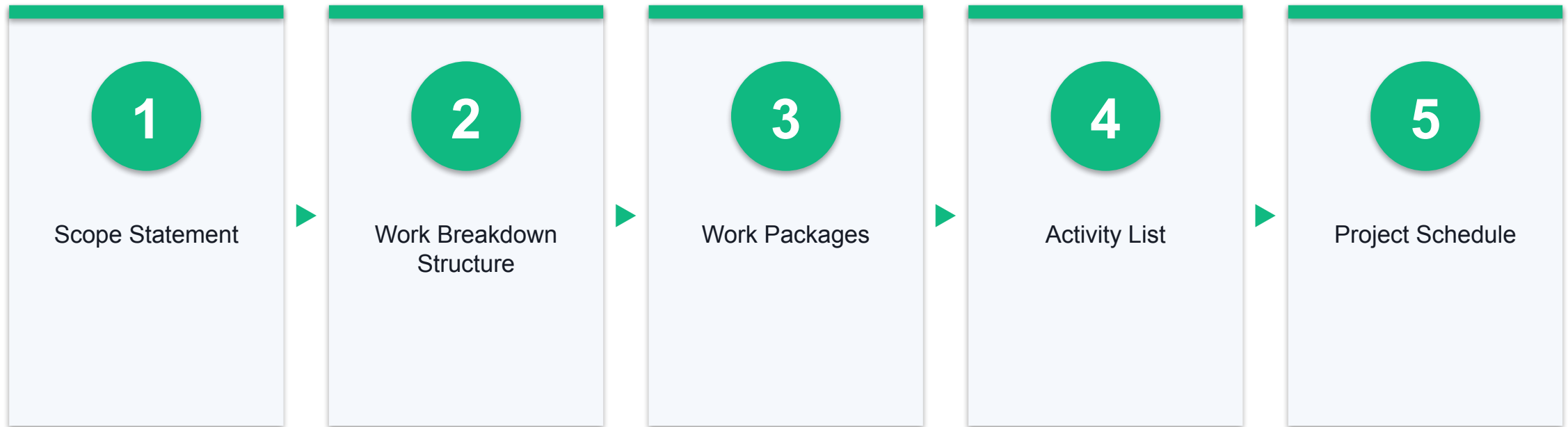
Additional scope information for each work package — usually a matrix.

4

The 100% rule

The children of any node must total all of that node's work, with no overlap.

From Scope to Schedule



WHAT EACH ENTRY HOLDS

The WBS Dictionary

1

WBS code identifier

2

Associated schedule activities

3

Resources required to complete the work

4

Cost estimates

5

Quality requirements

6

Acceptance criteria and technical references

Decompose Scope into a WBS and WBS Dictionary with GenAI

ACTIVITY 8

The WBS is the backbone of the plan — everything downstream, schedule and cost, is built from it. GenAI decomposes work quickly; your job is to check the decomposition is complete and mutually exclusive.

You'll produce

A three-level WBS and a WBS Dictionary for your work packages.

Tools: GenAI tool · WBS Template.pptx · WBS Dictionary.xlsx

Step-by-step instructions: [see your Learner Guide](#)

Decompose Scope into a WBS and WBS Dictionary with GenAI

ACTIVITY 8

PROMPT GenAI WITH

Create a three-level work breakdown structure for implementing an internal knowledge portal for a consulting firm. Make every lowest-level item a deliverable-based work package.

⚠ THEN CHECK IT

GenAI gives you a fast first draft. Check every figure, source and assumption before you use it.

Decompose Scope into a WBS and WBS Dictionary with GenAI

Check your work

Your WBS decomposes to assignable work packages, satisfies the 100% rule, includes PM/testing/training effort, and every dictionary entry has testable acceptance criteria.



SCHEDULE

Build the Schedule & Find the Critical Path

The Schedule Management Plan

1

Defines activities

How activities will be defined and progressively elaborated.

2

Names the method and tool

The scheduling methodology and the software used.

3

Sets the format

How the schedule will be presented.

4

Sets control criteria

How the schedule will be developed, controlled and updated during execution.

Schedule Presentation Formats

1

Roadmap

High-level view of direction over time — for executives.

2

Gantt Chart

Bars against a calendar showing activities, durations and overlap.

3

Milestone Chart

Key dates only — no durations.

4

Network Diagram

Activities and their dependencies — this is what reveals the critical path.

Activity Dependency Types

Type	Meaning	What the PM can do
Mandatory	Contractually required or inherent in the nature of the work	Must schedule it — there is no way around this sequence
Discretionary	Established from best practice or a preferred sequence	Can be changed if a better sequence exists or compression is needed
External	Activities performed outside the project team's work	Limited or no control — manage as a risk
Internal	Within project work, contingent on the team's own inputs	Full control

Leads and Lags

Lead — pull forward

- Lead
 - The amount of time an activity can be brought FORWARD
 - Allows a successor to start before the predecessor finishes
 - Written as a negative value, e.g. FS – 2 days

Lag — push back

- Lag
 - The amount of time an activity is DELAYED
 - Waiting time between activities — concrete curing, approval periods
 - Written as a positive value, e.g. FS + 3 days

Estimating Durations

1

Expert judgement

Ask the people who will actually do the work.

2

Analogous

Use a similar past activity — fast, lower accuracy.

3

Parametric

Use a rate — \$55 per metre, 8 hours per test case. Medium effort and accuracy.

4

Bottom-up

Add up the smallest pieces — high effort, high accuracy.

5

Three-point

$(O + M + P) / 3$ — useful when opinions differ.

6

PERT / Beta

$(O + 4M + P) / 6$ — weights the most likely estimate.

Wideband Delphi and Planning Poker

- The people who will do the work estimate the effort independently.
- The highest and lowest estimates explain their reasoning to the group.
- Everyone re-estimates. Repeat until the estimates converge.
- Useful in complex situations, and it surfaces hidden assumptions that a single estimate hides.

Schedule Definitions You Must Know

1

Free Float

How long an activity can be delayed without delaying its successor.

2

Total Float

How long an activity can be delayed without delaying the project.

3

Critical Path

The sequence of activities that determines the shortest possible project duration.

4

Forward & Backward Pass

The calculation that determines each activity's float.

THE CRITICAL PATH

Activities on the critical path have zero float.

Delay any one of them and you delay the whole project. This is why you must know which activities are critical — and GenAI's arithmetic on this must be checked.

Build the Schedule and Find the Critical Path with GenAI

ACTIVITY 9

You will turn your work packages into a sequenced, estimated schedule, draw the network diagram, and compute the critical path — then check whether GenAI's arithmetic can be trusted.

You'll produce

A Gantt chart, a schedule network diagram and a calculated critical path.

Tools: GenAI tool · Gantt Chart Template.xlsx · Schedule Network Diagram.xlsx

Step-by-step instructions: [see your Learner Guide](#)

Build the Schedule and Find the Critical Path with GenAI

ACTIVITY 9

PROMPT GenAI WITH

Using the task data provided, propose a project schedule for building a CRM system. Preserve the stated dependencies, identify the critical path and show your forward-pass and backward-pass calculations.

⚠ THEN CHECK IT

GenAI gives you a fast first draft. Check every figure, source and assumption before you use it.

Build the Schedule and Find the Critical Path with GenAI

Check your work

You can state the critical path, the total duration and the float on at least one non-critical activity, and you have checked GenAI's computed path against your own.



COST

Estimate Costs & Determine the Budget

Estimating Costs — Four Techniques

1

Analogous

Use historical information and expert judgement from a previous similar project. Low cost, reasonably accurate.

2

Parametric

Take a rate from historical data and apply it: $\text{cost} = \text{quantity} \times \text{rate per unit}$.

3

Bottom-up

Roll costs up from the smallest activity. Most accurate, but time consuming and costly.

4

Three-point

$C_e = (C_o + 4C_m + C_p) / 6$ — a weighted average that reduces bias and uncertainty.

Budgets — Predictive vs Agile

Predictive

- Predictive projects
 - Total budget built bottom-up from work package estimates
 - Contingency reserves added for identified risks
 - Management reserves added for unforeseen scope
 - Once approved, changed only by change request

Agile

- Agile projects
 - The team is fully funded for a set period
 - Cost and time are FIXED
 - Scope is the variable — prioritised by the product owner
 - Value is delivered incrementally within the funded period

Reserves

1

Contingency Reserve

Set aside for identified risks should they occur. Sits INSIDE the cost baseline. Controlled by the PM.

2

Management Reserve

Set aside for unforeseen work within scope. Sits OUTSIDE the cost baseline. Requires management approval.

3

Reserve Analysis

Compares the amount of risk remaining against the amount of contingency remaining.

4

Cost Baseline

All work package costs (bottom-up) plus the contingency reserve.

Estimate Costs and Build the Budget with GenAI

ACTIVITY 10

You will apply the four estimating techniques to the same work, compare what each produces, and assemble a cost baseline with contingency and management reserves.

You'll produce

A cost baseline built bottom-up from your WBS, with reserves and a documented estimating basis.

Tools: GenAI tool · Cost Benefit Analysis.xlsx · your WBS from Activity 8

Step-by-step instructions: [see your Learner Guide](#)

Estimate Costs and Build the Budget with GenAI

ACTIVITY 10

PROMPT GenAI WITH

Estimate the cost of each work package in this WBS using bottom-up estimating. Show assumed rates and quantities separately from totals, then add contingency and management reserves without double counting.

⚠️ THEN CHECK IT

GenAI gives you a fast first draft. Check every figure, source and assumption before you use it.

Estimate Costs and Build the Budget with GenAI

Check your work

You have one cost baseline built bottom-up, four comparable estimates for the same work, no unmarked assumed rates, and a stated contingency and management reserve you can justify.

Recap — Project Initiation and Planning with GenAI

- Apply structured prompting patterns to produce usable project management output.
- Establish the economic feasibility of a project using GenAI-assisted analysis.
- Develop project parameters with GenAI according to business requirements.
- Identify stakeholders and plan engagement using GenAI-assisted analysis.
- Allocate project responsibilities unambiguously using a responsibility assignment matrix.
- Elicit, structure and prioritise requirements using GenAI-supported techniques.
- Trace every requirement from source through to test and deliverable.

Lunch Break

1 hour

TOPIC 02

Project Execution and Delivery with GenAI

Directing work · knowledge · quality · resources · team · communications



Key Concepts — Topic 02

1

Direct and manage project work

Turn the plan into deliverables; distinguish work performance data, information and reports.

2

Manage project knowledge

Capture explicit knowledge and surface tacit knowledge; GenAI summarises lessons learned as you go.

3

Quality management

Cost of quality, PDCA, DMAIC, Kaizen; fishbone, five whys and Pareto for root-cause analysis.

4

Acquire and develop resources

Resource levelling vs smoothing; Tuckman's ladder; motivational theories.

5

Manage the team

Conflict management approaches, emotional intelligence and interpersonal skills.

6

Manage communications

The $N(N-1)/2$ channel formula, the Five Cs, push vs pull, and barriers to communication.

7

Optimising resources and budget

Use GenAI to model resource and budget options, then justify the trade-off you choose.

GenAI Activities — Topic 02

Activities 11–13

- Direct Project Work and Report Performance with GenAI
- Manage Project Knowledge with GenAI
- Root Cause Analysis with GenAI — Fishbone, Five Whys and Pareto

Activities 14–16

- Continuous Improvement with GenAI — PDCA and DMAIC
- Optimise Resources and Budget with GenAI
- Develop and Manage the Team with GenAI

Activities 17–17

- Plan and Manage Communications with GenAI



EXECUTION

Direct and Manage Project Work

Directing the Work

1

Implement the plan

Turn the project management plan into actual deliverables.

2

Manage the flow

Keep work moving and remove impediments for the team.

3

Implement approved changes

Only approved changes — everything else goes through change control.

4

Collect the data

Work performance data flows out of the work and into your reporting.

Data → Information → Reports

Work Performance DATA

- The raw observations
- '14 defects open'
- 'Actual cost \$62,000'
- No meaning on its own

Work Performance INFORMATION

- Comparison of two or more data points
- 'Schedule variance is –\$10,000'
- '8 work packages done vs 10 planned'
- Now it means something

Work Performance REPORTS

- The physical or electronic representation
- Status reports, dashboards, memos
- Intended to raise action or awareness
- What the sponsor actually sees

Managing Project Information

1

PMIS

The Project Management Information System gathers, integrates and shares project data.

2

Artifact management

Systems that store and maintain project artifacts.

3

Version control

Every update gets a new version number, a date/time stamp and the name of who changed it.

4

Configuration management

Controls which version of a deliverable is the approved one.

Direct Project Work and Report Performance with GenAI

ACTIVITY 11

Executing a project generates a flood of raw data. You will use GenAI to turn that data into information, and information into a report a sponsor will actually read — while learning the distinction that PMBOK draws between the three.

You'll produce

A project status report generated from raw work performance data.

Tools: GenAI tool · Project Manager Buddy GPT

Step-by-step instructions: [see your Learner Guide](#)

Direct Project Work and Report Performance with GenAI

ACTIVITY 11

PROMPT GenAI WITH

Here is raw work performance data from my project [paste]. Compare it with the plan, distinguish data from interpretation, and produce a one-page status report for an executive sponsor.

⚠ THEN CHECK IT

GenAI gives you a fast first draft. Check every figure, source and assumption before you use it.

Direct Project Work and Report Performance with GenAI

Check your work

Your report distinguishes data from information, every comparison uses a real baseline figure, and it opens with status, biggest problem and the decision required.

Manage Project Knowledge

1

Explicit knowledge

Documented and easily shared — plans, reports, procedures. Archive it.

2

Tacit knowledge

Held in people's heads — experience, judgement, relationships. Must be surfaced.

3

Curate as an asset

Treat knowledge as an asset to the team and the organisation.

4

Lessons learned

Captured continuously through the project, not only at the end.

Manage Project Knowledge with GenAI

ACTIVITY 12

Knowledge that lives only in people's heads leaves when they do. You will use GenAI to turn messy project conversations into a structured, searchable knowledge asset and a lessons-learned register.

You'll produce

A lessons-learned register and a knowledge asset entry for your project.

Tools: GenAI tool · Claude (long-document handling)

Step-by-step instructions: [see your Learner Guide](#)

Manage Project Knowledge with GenAI

ACTIVITY 12

PROMPT GenAI WITH

From these project records [paste], extract decisions and owners, open questions, lessons learned, and tacit knowledge that is implied but not stated. Quote the evidence used for each extraction.

⚠ THEN CHECK IT

GenAI gives you a fast first draft. Check every figure, source and assumption before you use it.

Manage Project Knowledge with GenAI

Check your work

Your register has actionable lessons with root causes, at least one item of genuinely tacit knowledge surfaced, and your group can state your organisation's data-confidentiality limits.



QUALITY

Manage and Control Quality

Quality in Predictive vs Agile Projects

Predictive

- Predictive
 - Product is developed, then tested
 - Testing happens in sequence, usually in large batches
 - Quality assurance is often a separate function

Agile

- Agile
 - Quality is everyone's responsibility
 - Correct requirements and acceptance criteria up front
 - Simple design and regular refactoring
 - Test-first: write the test, then the code that passes it
 - Small deliverable demonstrated in the sprint review

WHY QUALITY IS PLANNED, NOT INSPECTED IN

Quality is more expensive to fix the closer it gets to your customer.

A defect caught in design costs almost nothing. The same defect caught by a customer costs you rework, reputation and trust.

The Cost of Quality

1

Prevention costs

Preventing future defects — training, prototypes, refactoring, good design.

2

Appraisal costs

Finding defects — testing, quality audits, inspections, the sprint review.

3

Internal failure

The product fails internally, usually during testing or review.

4

External failure

Non-conformance reaches the customer — complaints, returns, reputational damage.

Quality Methods for Continuous Improvement

1

Six Sigma / DMAIC

Define, Measure, Analyse, Improve, Control — focused on removing variation and waste.

2

Kaizen

'Change for better' — continuous small improvements by everyone.

3

PDCA

Plan-Do-Check-Act — the Shewhart / Deming cycle.

4

Agile methods

Scrum, Kanban and their built-in inspect-and-adapt loops.

Root Cause Analysis Tools

1

Ishikawa / Fishbone

Cause-and-effect diagram grouping causes into categories.

2

The Five Whys

Ask 'why' repeatedly until you reach something you can actually change.

3

Pareto Chart

The 80/20 rule — find the vital few causes creating most of the problem.

4

Six Thinking Hats

Examine the problem deliberately from six different perspectives.

Root Cause Analysis with GenAI — Fishbone, Five Whys and Pareto

ACTIVITY 13

When delivery goes wrong, the visible problem is rarely the real one. You will run three complementary root-cause techniques on the same problem with GenAI's help and compare what each reveals.

You'll produce

A fishbone diagram, a five-whys chain and a Pareto analysis of the same delivery problem.

Tools: GenAI tool · Fishbone Diagram Template.pptx · Pareto Principle Template.xlsx

Step-by-step instructions: [see your Learner Guide](#)

Root Cause Analysis with GenAI — Fishbone, Five Whys and Pareto

ACTIVITY 13

PROMPT GenAI WITH

Analyse the issue records provided and identify the most likely root causes of recurring project deadline misses. Produce a fishbone structure, a five-whys chain and a Pareto-ready category table without inventing evidence.

⚠️ THEN CHECK IT

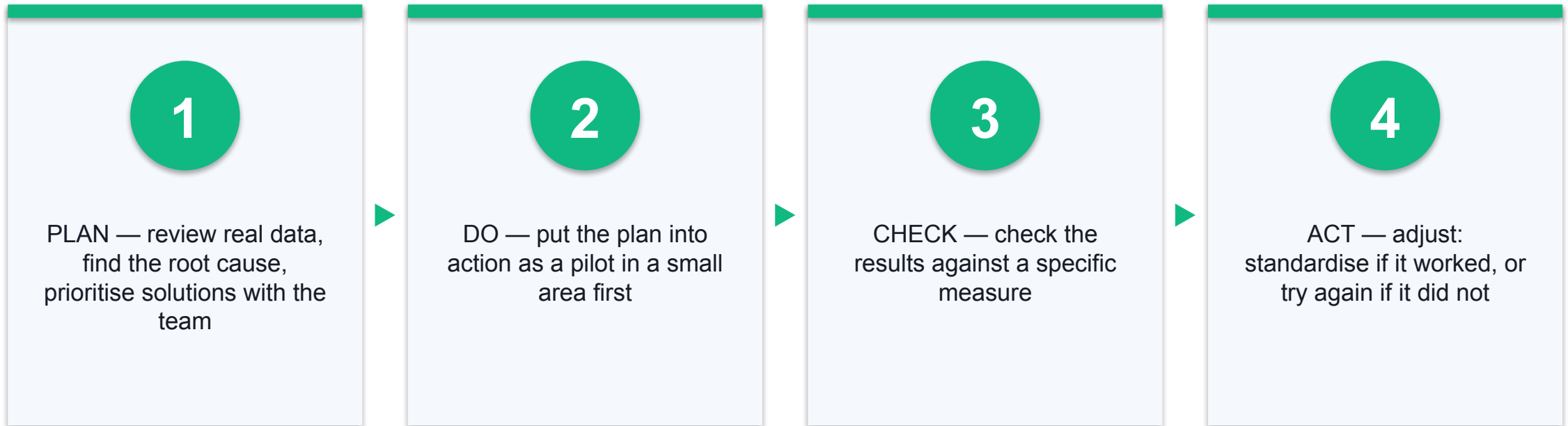
GenAI gives you a fast first draft. Check every figure, source and assumption before you use it.

Root Cause Analysis with GenAI — Fishbone, Five Whys and Pareto

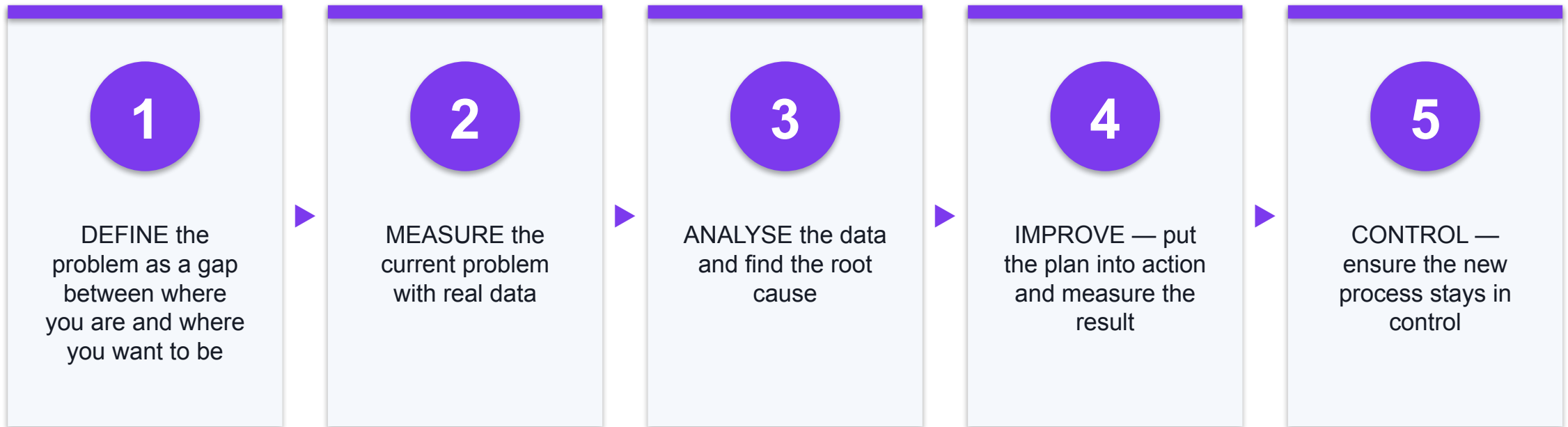
Check your work

You have all three artifacts for one problem, you added at least two organisation-specific causes the AI missed, and your five-whys chain ends at something you can change.

PDCA — The Deming Cycle



DMAIC — Six Sigma



Continuous Improvement with GenAI — PDCA and DMAIC

ACTIVITY 14

Having found a root cause, you need a disciplined way to fix it and make the fix stick. You will run the same improvement through both PDCA and DMAIC and see where they differ.

You'll produce

A completed PDCA cycle and a DMAIC project definition for your improvement.

Tools: GenAI tool · Plan Do Check Act.pptx · DMAIC.pptx

Step-by-step instructions: see your Learner Guide

Continuous Improvement with GenAI — PDCA and DMAIC

ACTIVITY 14

PROMPT GenAI WITH

Design a PDCA cycle to address [root cause]. For Plan, state the evidence and prioritised solution; for Do, define a small pilot; for Check, define the success measure; for Act, state what happens if the pilot succeeds or fails.

⚠️ THEN CHECK IT

GenAI gives you a fast first draft. Check every figure, source and assumption before you use it.

Continuous Improvement with GenAI — PDCA and DMAIC

Check your work

Your Check/Control stages specify a number and a date, your Do stage is a pilot, and you can justify choosing PDCA or DMAIC for this problem.

Compliance Requirements

1

Know the standards

Government regulations, organisational policies, product and project quality requirements.

2

Classify

Determine which compliance categories apply to your project.

3

Assess threats

Determine potential threats to compliance and analyse the consequences of non-compliance.

4

Act

Determine the necessary approach and actions to address compliance needs.



RESOURCES

Acquire, Develop and Manage Resources

The Resource Management Plan

1

Identify resources

Both people and physical resources, and how to acquire them.

2

Roles and responsibilities

Role (function), Authority (rights to decide and use resources),
Responsibility (assigned duty).

3

Competencies

The skills required, and the training strategy to close any gaps.

4

Organisation chart

A visual showing resource categories and reporting relationships.

5

Resource controls

Ensuring non-human resources are available when needed.

6

Recognition plan

How contribution will be recognised and rewarded.

Resource Optimisation

Levelling

- Resource Levelling
 - Used when a shared resource is over-allocated
 - Or assigned to two or more activities in the same period
 - Adjusts the schedule to resolve it
 - MAY change the critical path and the end date

Smoothing

- Resource Smoothing
 - Adjusts activities only within their free and total float
 - The critical path is NOT changed
 - The end date is preserved
 - May not fully resolve the over-allocation

Schedule Compression

Crashing — spend money

- Crashing
 - Add resources to shorten duration for least incremental cost
 - Overtime, extra people, paying to expedite
 - ONLY works on critical path activities
 - Increases cost, and often risk

Fast tracking — take risk

- Fast Tracking
 - Perform activities in parallel that were planned in sequence
 - Overlaps critical path activities to shorten duration
 - No additional cost up front
 - Increases risk of rework — which can cost more later

Optimise Resources and Budget with GenAI

ACTIVITY 15

This is the heart of LO2. A resource is over-allocated and the budget is under pressure — you will use GenAI to generate genuine alternatives, evaluate the trade-offs, and defend the option you choose.

You'll produce

A resource optimisation decision with a documented trade-off analysis.

Tools: GenAI tool · your schedule from Activity 9 · Gantt Chart Template.xlsx

Step-by-step instructions: [see your Learner Guide](#)

Optimise Resources and Budget with GenAI

ACTIVITY 15

PROMPT GenAI WITH

I have an over-allocated resource on these activities [paste] and a 15% budget reduction. Compare resource levelling, resource smoothing, crashing and fast tracking, including cost, schedule, quality and risk trade-offs.

⚠ THEN CHECK IT

GenAI gives you a fast first draft. Check every figure, source and assumption before you use it.

Optimise Resources and Budget with GenAI

Check your work

Your chosen option is correctly named, crashing (if used) targets the critical path, and you can justify the cost/schedule/risk trade-off in three sentences.



TEAM

Develop and Manage the Project Team

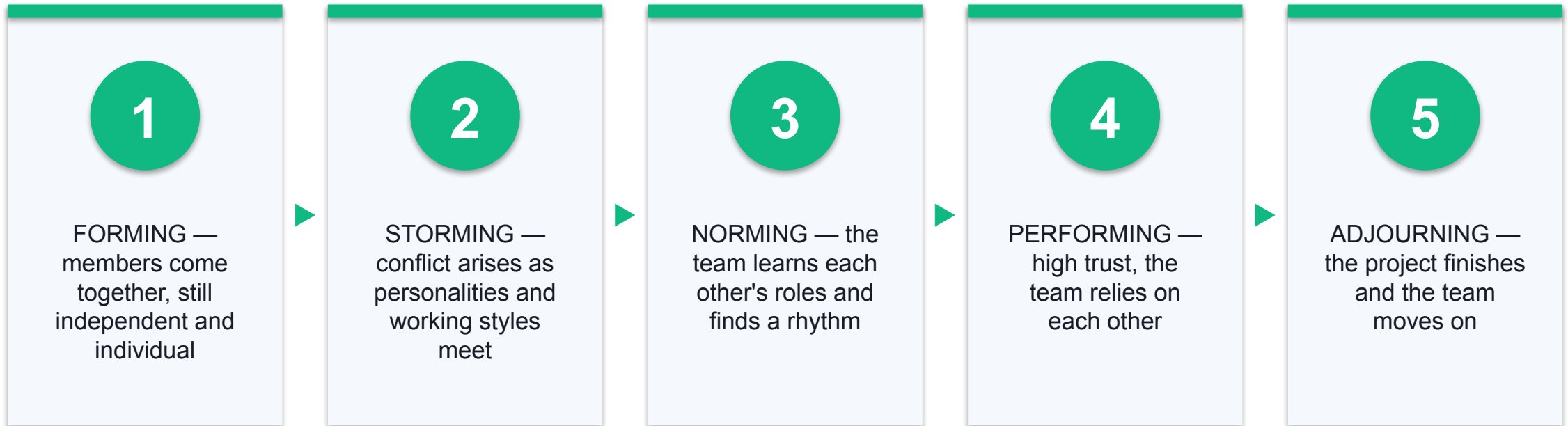
AN EFFECTIVE TEAM IS THE KEY TO SUCCESS



Of all the things I've done, the most vital is coordinating the talents of those who work for us and pointing them toward a certain goal.

Walt Disney

Tuckman's Ladder



Team Building

1

Cohesion drives performance

Cohesion and solidarity help teams perform better.

2

Leadership facilitates it

Good leadership helps bonding between team members.

3

Builds trust and empathy

Activities build unity, trust and focus on the team over the individual.

4

Formal or informal

Brief or extended, facilitated by you or by a professional.

Motivational Theories

1

Maslow's Hierarchy of Needs

Physiological, safety, belonging, esteem, self-actualisation — lower needs first.

2

Herzberg's Motivation-Hygiene

Hygiene factors prevent dissatisfaction; motivators create satisfaction.

3

McGregor's Theory X and Y

X: people avoid work and need direction. Y: people take pride in work and need trust.

4

McClelland's Achievement Motivation

Needs for achievement, affiliation or power — one becomes dominant.

Conflict Management

1

Everyone's responsibility

All team members and stakeholders are responsible for managing conflict.

2

The PM influences it

Through interpersonal skills and servant leadership.

3

The team is empowered

To resolve conflicts themselves; the lead facilitates resolution.

4

Conflict is not all bad

Disruption and innovation are connected — encourage exchange and disagreement, prevent escalation.

Causes of Conflict

1

Competition for scarce resources

2

Differences in objectives, values and perceptions

3

Disagreements about roles and work activities

4

Communication breakdowns

5

Projects are unique — the team has not worked together before

6

Ambiguous accountability

Five Conflict Management Approaches

Approach	What it means	When to use it
Collaborate / Problem Solve	Win/Win — incorporate multiple viewpoints, seek consensus	When you need commitment and have time; the ideal default
Compromise / Reconcile	Lose/Lose — each side gives something up	When both parties have equal power and time is short
Smooth / Accommodate	Lose/Win — emphasise agreement, concede your position	When the relationship matters more than the issue
Force / Direct	Win/Lose — pursue your view at others' expense	In an emergency, or on safety and deadline-critical decisions
Withdraw / Avoid	Retreat and postpone the issue	When the issue is trivial, or emotions need to cool first

Emotional Intelligence

1

Self-Awareness

Understanding how your actions affect others.

2

Self-Management

Thinking before you act.

3

Social Awareness

Being empathetic to those around you.

4

Social Skill

Establishing rapport and managing your own attitude.

Develop and Manage the Team with GenAI

ACTIVITY 16

Delivery problems are usually people problems. You will use GenAI to prepare for a difficult conversation and to select a conflict management approach — while judging where AI advice on people is genuinely useful and where it is not.

You'll produce

A prepared difficult conversation and a team development plan mapped to Tuckman's stages.

Tools: GenAI tool · Claude · Scrum Agile Transformation Studio (alfredang.github.io/scrum)

Step-by-step instructions: see your **Learner Guide**

Develop and Manage the Team with GenAI

ACTIVITY 16

PROMPT GenAI WITH

My project team is in the [stage] stage of Tuckman's model and showing these behaviours [paste]. Recommend leadership actions that can move the team forward, and explain which observations support each action.

⚠ THEN CHECK IT

GenAI gives you a fast first draft. Check every figure, source and assumption before you use it.

Develop and Manage the Team with GenAI

Check your work

You have a conversation plan you would actually use, a named conflict approach with justification, an exported retrospective file, and a clear line on what not to delegate to AI.



COMMUNICATIONS

Plan and Manage Communications

THE COMMUNICATION CHANNEL FORMULA

$$N \times (N - 1) \div 2$$

A team of 10 people has 45 communication channels. A team of 15 has 105. Communication complexity grows faster than the team.

The Communications Management Plan

1

Who

Senders, receivers, and who authorises confidential information.

2

What

The type of information and the reason for communicating it.

3

How

Language, format, content and level of detail.

4

When

Time frame and frequency; whether acknowledgment is required.

5

Escalation

The process and guidance for escalating issues.

6

Resources

Methods and technologies used, plus allocated time and budget.

The Five Cs of Communication

1

Correct

Correct grammar and spelling — errors undermine your credibility.

2

Concise

Eliminate excess words. Respect the reader's time.

3

Clear

A clear purpose, directed at the needs of the reader.

4

Coherent

A logical flow of ideas.

5

Controlling

Control the flow of words and ideas — lead with what matters.

Push vs Pull Communication

Push

- Push
 - Sent directly to specific recipients
 - Emails, reports, memos, voicemails
 - Ensures it was DISTRIBUTED — not that it was understood
 - Use for what must be read

Pull

- Pull
 - Recipients access it at their own discretion
 - Intranet sites, dashboards, knowledge repositories
 - Suits large volumes and large audiences
 - Use for reference information

Communication Skills

1

Active listening

Stay engaged, show interest, summarise back to check understanding.

2

Feedback

Written, verbal or non-verbal — ensures the message was heard correctly.

3

Non-verbal

Body language, gestures, tone of voice, facial expressions.

4

Presentations

Formal delivery of progress and updates to stakeholders.

5

Cultural awareness

Habits, priorities, celebrations and ways of working differ.

6

Managing expectations

Set expectations early; negotiate to manage them continuously.

Barriers to Communication

1

Language differences

2

Cultural differences

3

Physical barriers — noise, distance, poor infrastructure

4

Psychological barriers — stress, emotion

5

Perceptual barriers — different interpretations

6

Information overload

7

Jargon and technical language

8

Attitudes, prejudices and poor listening

Communication Styles

1

Assertive

Direct, respectful and confident — the style to aim for.

2

Aggressive

Dominating, confrontational and disrespectful.

3

Passive

Submissive, avoids confrontation, indirect.

4

Passive-aggressive

Indirect, with disguised hostility.

5

Manipulative

Controlling and deceitful, using influence to achieve hidden goals.

Plan and Manage Communications with GenAI

ACTIVITY 17

You will calculate how communication complexity grows, then use GenAI to build a communications plan and adapt a single message for very different audiences.

You'll produce

A communications management plan and the same update written for three audiences.

Tools: GenAI tool · your stakeholder register from Activity 4

Step-by-step instructions: [see your Learner Guide](#)

Plan and Manage Communications with GenAI

ACTIVITY 17

PROMPT GenAI WITH

Build a communications management plan from the stakeholder data provided. For each stakeholder specify information needed, purpose, format, level of detail, frequency, channel and sender.

⚠ THEN CHECK IT

GenAI gives you a fast first draft. Check every figure, source and assumption before you use it.

Plan and Manage Communications with GenAI

Check your work

Your plan gives each stakeholder a specific frequency and channel, uses push and pull deliberately, and your three message versions genuinely differ in what they lead with.

Recap — Project Execution and Delivery with GenAI

- Turn raw project data into work performance information and reports.
- Capture explicit and tacit knowledge so it is reusable by the organisation.
- Diagnose the root cause of a delivery problem, not just its symptoms.
- Structure an improvement cycle and control the result.
- Resolve over-allocation and budget pressure by modelling options with GenAI (K3).
- Advise staff in resolving challenges during project delivery (A2).
- Build a communications plan that matches each stakeholder's real needs.

Tea Break

15 minutes

TOPIC 03

Managing Project Risks with GenAI

Risk · procurement · monitoring & control ·
earned value · closure



Key Concepts — Topic 03

1

Risk vs issue

Risks are future and may be positive or negative (risk register); issues are present and always negative (issue log).

2

Identify and analyse risk

RBS, brainstorming, SWOT and the Delphi technique; qualitative probability/impact then quantitative analysis.

3

Quantitative analysis

Sensitivity (tornado) charts and decision-tree expected monetary value.

4

Plan risk responses

Avoid, transfer, mitigate, accept — escalate, exploit, share, enhance; secondary and residual risk.

5

Procurement

RFI/RFP/RFQ, contract types, claims and alternative dispute resolution.

6

Monitor and control

Integrated change control, validate and control scope, schedule and cost.

7

Earned value analysis

PV, EV, AC; CV, SV, CPI, SPI; EAC and ETC forecasting.

8

Control quality & close

Check sheets, control charts and the rule of seven; then close the project and capture lessons learned.

GenAI Activities — Topic 03

Activities 18–20

- Identify Project Risks with GenAI
- Qualitative Risk Analysis with GenAI
- Quantitative Risk Analysis with GenAI — Tornado Charts and Decision Trees

Activities 21–23

- Plan Risk Responses with GenAI
- Plan Procurement and Manage Contracts with GenAI
- Earned Value Analysis with GenAI

Activities 24–25

- Control Quality with GenAI — Check Sheets and Control Charts
- Monitor, Control Change and Close the Project with GenAI



RISK

Identify and Analyse Project Risks

DEFINITION

What is Project Risk?

1

Known and unknown causes

Risk originates inside and outside the business environment.

2

Signalled by a trigger

A trigger condition indicates the risk is developing.

3

Positive or negative

Risks can be opportunities as well as threats.

4

Becomes an issue

If a risk materialises it becomes an issue, and you must act.

Risks vs Issues

Risk

- RISKS
 - Focused on the FUTURE
 - Can be positive or negative
 - Documented in the RISK REGISTER
 - The response is called a 'risk response'
 - You plan for them in advance

Issue

- ISSUES
 - Focused on the PRESENT
 - Always negative
 - Documented in the ISSUE LOG
 - The response is called a 'workaround'
 - You deal with them now

WHY YOU PLAN FOR RISK

Risk, like quality, is cheaper to fix early.

A risk identified in planning can be designed out. The same risk discovered in execution costs you schedule, budget and credibility.

Risk Identification Techniques

1

Risk Breakdown Structure

A hierarchy of risk categories used as a checklist when brainstorming.

2

Brainstorming

Generate risks freely with the team before filtering.

3

Nominal group technique

Individuals rank risks privately, then scores are tallied.

4

SWOT analysis

Strengths, Weaknesses, Opportunities, Threats.

5

Assumption analysis

Every assumption in the plan is a risk if it proves false.

6

Delphi technique

Anonymous expert input, iterated until consensus emerges.

Common Types of Project Risk

1

Budget, cost and exchange rate

2

Schedule and project dependencies

3

Scope creep and requirement risk

4

Technology, design and architecture

5

Supplier, procurement and contract

6

Regulatory, legal and compliance

7

Resource, skill and sponsor support

8

Quality, security and reputation

CAUSE → EVENT → EFFECT

Writing a Proper Risk Statement

- A risk is not a single word. 'Vendor' is not a risk.
- Use the structure: CAUSE → RISK EVENT → EFFECT.
- 'Because the vendor has only one integration engineer, their resource may be unavailable in December, delaying UAT by two weeks.'
- Now it is specific enough to assign an owner, a trigger and a response.

Identify Project Risks with GenAI

ACTIVITY 18

GenAI is genuinely excellent at risk identification — it will suggest risks your team would never think of. It is correspondingly poor at knowing which of them matter to you. You will use both facts.

You'll produce

A populated risk register with risks sourced from GenAI and from your team.

Tools: GenAI tool · Project Manager Buddy GPT · Risk Register template

Step-by-step instructions: [see your Learner Guide](#)

Identify Project Risks with GenAI

ACTIVITY 18

PROMPT GenAI WITH

Act as an expert project manager in the [industry] working on a [project type]. Use the supplied project facts to draft a risk register covering cause, event, effect, category, owner and evidence. Separate facts from assumptions.

⚠ THEN CHECK IT

GenAI gives you a fast first draft. Check every figure, source and assumption before you use it.

Identify Project Risks with GenAI

Check your work

Your register has risks from both sources, every entry is written as cause → event → effect with an owner, and you can name one risk your team found that the AI missed.

Qualitative Risk Analysis

1

Do it first

Subjective assessment comes before any quantitative work.

2

Probability × Impact

Combine the probability of occurrence with the impact if it occurs.

3

Prioritise

The output is a prioritised shortlist, not a score for every risk.

4

Then quantify

Move to quantitative analysis only if further support is required.

Other Risk Factors

1

Urgency

How soon a response is needed to be effective.

2

Proximity

How close the risk is to occurring.

3

Dormancy

The period between the risk occurring and you discovering it.

4

Manageability

How easily the risk can be managed if it occurs.

5

Controllability

How easily the risk outcome can be controlled.

6

Strategic impact

How the risk affects the organisation's goals.

7

Connectivity

How related this risk is to other project risks.

8

Propinquity

The degree to which stakeholders actually care about the risk.

The Risk Register

1

Risk ID and title

2

Category and description (cause → event → effect)

3

Probability, impact and score

4

Risk owner

5

Planned response and trigger condition

6

Current status and residual risk

Qualitative Risk Analysis with GenAI

ACTIVITY 19

You cannot act on fifty risks. Qualitative analysis prioritises them so you spend your attention where it matters — and you will test how consistent GenAI's scoring actually is.

You'll produce

A scored probability/impact matrix with a prioritised risk shortlist.

Tools: GenAI tool · Probability and Impact Matrix template

Step-by-step instructions: [see your Learner Guide](#)

Qualitative Risk Analysis with GenAI

ACTIVITY 19

PROMPT GenAI WITH

Score each supplied risk for probability and impact from 1 to 5, calculate the risk score, place it in a probability-impact matrix and explain any score that is not directly supported by the data.

⚠️ THEN CHECK IT

GenAI gives you a fast first draft. Check every figure, source and assumption before you use it.

Qualitative Risk Analysis with GenAI

Check your work

Your top ten risks have group-agreed scores, your top five carry proximity and dormancy assessments, and you can explain why AI scoring alone is not sufficient.

Quantitative Risk Analysis

Tornado charts

- Sensitivity Analysis / Tornado Chart
 - Shows which variables most affect the outcome
 - Each bar is a variable's high-to-low impact range
 - Longest bars at the top — those are your priorities
 - Tells you where to focus management attention

Decision trees

- Decision Tree Analysis
 - Shows decisions, outcomes and their probabilities
 - Each path carries a cost and a payoff
 - Produces an Expected Monetary Value per branch
 - $EMV = \sum (\text{probability} \times \text{payoff}) - \text{branch cost}$

Quantitative Risk Analysis with GenAI — Tornado Charts and Decision Trees

ACTIVITY 20

When a decision carries real money, qualitative scoring is not enough. You will run a sensitivity analysis and a decision tree, using GenAI for the setup and doing the arithmetic yourself.

You'll produce

A tornado chart identifying your most sensitive risk variables and a decision tree with expected monetary values.

Tools: GenAI tool · Tornado Chart - Sensitivity Analysis.xlsx · Decision Tree Analysis.xlsx

Step-by-step instructions: [see your Learner Guide](#)

Quantitative Risk Analysis with GenAI — Tornado Charts and Decision Trees

ACTIVITY 20

PROMPT GenAI WITH

Use the supplied high, base and low values to rank the ten variables by their effect on project cost or schedule. Produce tornado-chart data and one decision tree with expected monetary values, showing every calculation.

⚠ THEN CHECK IT

GenAI gives you a fast first draft. Check every figure, source and assumption before you use it.

Quantitative Risk Analysis with GenAI — Tornado Charts and Decision Trees

Check your work

Your tornado chart is ordered by sensitivity with justified asymmetric ranges, and you have hand-verified the EMV of every decision tree branch.

Risk Response Strategies

Threats

- For THREATS (negative risk)
 - AVOID — eliminate the cause; change the plan
 - TRANSFER — shift the impact to a third party (insurance, contract)
 - MITIGATE — reduce the probability or the impact
 - ACCEPT — take no action; hold a contingency reserve

Opportunities

- For OPPORTUNITIES (positive risk)
 - EXPLOIT — make certain the opportunity happens
 - SHARE — partner with someone better able to capture it
 - ENHANCE — increase the probability or the impact
 - ACCEPT — take it if it arrives, but do nothing to cause it

After the Response — What's Left

1

Trigger condition

The observable signal that the risk is developing. A response with no trigger is never implemented in time.

2

Secondary risk

A NEW risk created directly by your response. Transferring risk to a vendor creates vendor dependency risk.

3

Residual risk

What REMAINS after the response is implemented. It must be accepted knowingly.

4

Fallback plan

What you do if the primary response fails. Funded by the contingency reserve.

Plan Risk Responses with GenAI

ACTIVITY 21

A risk register with no responses is a list of things you will be surprised by. You will plan a response for every priority risk and check for the second-order effects that responses create.

You'll produce

A response plan for every priority risk, with owners, triggers and fallback plans.

Tools: GenAI tool · your risk register from Activities 18-19

Step-by-step instructions: [see your Learner Guide](#)

Plan Risk Responses with GenAI

ACTIVITY 21

PROMPT GenAI WITH

For each priority risk [paste], recommend one primary response strategy from avoid, transfer, mitigate, accept, escalate, exploit, share or enhance. Give the action, owner, trigger, fallback and response cost.

⚠️ THEN CHECK IT

GenAI gives you a fast first draft. Check every figure, source and assumption before you use it.

Plan Risk Responses with GenAI

Check your work

Every priority risk has a correctly-named strategy, an owner, an observable trigger, an identified secondary and residual risk, and your top three have fallback plans.



PROCUREMENT

Plan Procurement and Manage Contracts

Planning the Procurement Strategy

1

Any external resource

Procurement covers vendors, consultants, contractors and equipment.

2

Agreement vs contract

You can have an agreement without a contract — but never a contract without an agreement.

3

Claims

Arise when buyer and seller cannot agree that a change occurred.

4

ADR

If a claim is unresolved, Alternative Dispute Resolution is used — as defined in the contract.

Bid Documents

1

RFI — Request for Information

When you need more information about what sellers can provide.

2

RFP — Request for Proposal

When the solution is not obvious and you need sellers to propose one.

3

RFQ — Request for Quote

When you know exactly what you want and need a price.

Contract Types — Who Carries the Risk

Contract type	Variants	Who carries cost-overflow risk / when to use
Fixed Price	Firm Fixed Price; FP + Incentive Fee; FP with Economic Adjustment	SELLER carries it. Use when scope is straightforward and requirements are well defined
Cost Reimbursable	Cost plus Fixed Fee; plus Incentive Fee; plus Award Fee	BUYER carries it. Use to incentivise the seller, or when work is expected to change
Time and Materials	Hybrid of both	Shared. Use when a precise statement of work is unavailable, or for staff augmentation

Components of a Contract

1

Description of work — deliverables and scope

2

Delivery date and schedule information

3

Responsibilities of both parties

4

Price and payment terms

5

Provisions for termination

6

Guarantees and warranties

7

Intellectual property

8

Security, confidentiality and data privacy

Managing Contract Disputes

1

Warranty

A promise, explicit or implied, that goods or services meet a predetermined standard.

2

Waiver

One party agrees to forfeit a claim without the other becoming liable.

3

Breach of contract

Failure to meet some or all obligations — may result in damages or litigation.

4

Cease and desist

A letter instructing a party to stop allegedly illegal activity.

Plan Procurement and Manage Contracts with GenAI

ACTIVITY 22

Procurement transfers risk — but only if you choose the right contract type. You will use GenAI to draft procurement documents and to reason about where risk actually sits in each contract type.

You'll produce

A procurement strategy with a justified contract type and a drafted RFP outline.

Tools: GenAI tool · Claude

Step-by-step instructions: [see your Learner Guide](#)

Plan Procurement and Manage Contracts with GenAI

ACTIVITY 22

PROMPT GenAI WITH

Compare Firm Fixed Price, Cost Plus Fixed Fee and Time and Materials for this procurement. State who carries cost-overrun risk, when each contract fits, and recommend one option using the supplied evaluation criteria.

⚠️ THEN CHECK IT

GenAI gives you a fast first draft. Check every figure, source and assumption before you use it.

Plan Procurement and Manage Contracts with GenAI

Check your work

You have the right bid document type, a contract type justified by who should carry scope risk, and a component checklist including IP, data privacy and dispute resolution.



MONITOR & CONTROL

Monitor Work, Control Change and Measure Performance

Monitoring and Controlling Project Work

1

Evaluate project progress against the baseline

2

Manage issues and impediments

3

Manage changes through integrated change control

4

Support project team performance

5

Implement ongoing improvements

6

Monitor the external business environment

Causes of Project Change

1

Inaccurate initial estimates

2

New or changed regulations

3

Missed requirements

4

Specification changes

5

Changes to the business case

6

External market or technology shifts

Types of Change Request

1

Corrective action

Realigns the performance of project work with the project management plan.

2

Preventive action

Ensures FUTURE performance aligns with the plan.

3

Defect repair

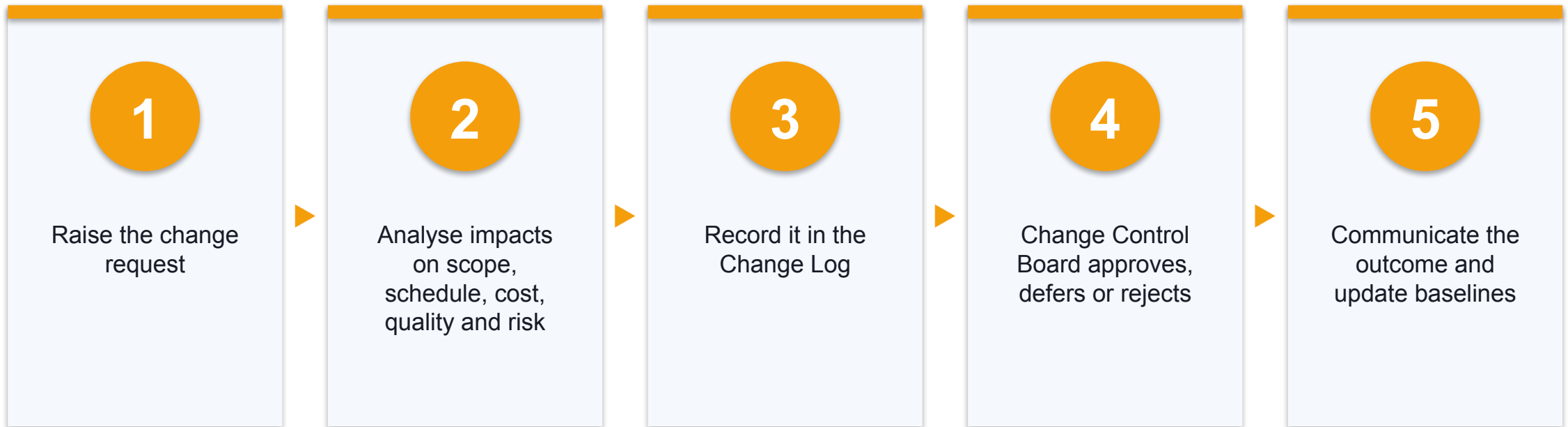
Modifies a non-conformance within the project.

4

Change

Modifies a project baseline — this is the one that needs formal approval.

Integrated Change Control



Reporting on Performance

1

Milestone schedule

High-level visualisation of progress against planned dates.

2

Quality reports

Charts and reports based on quality metrics.

3

Earned value reports

Graphs and values based on the EVM equations.

4

Variance analysis

Comparing actual against expected results.

5

Work performance reports

The formal status representation.

6

Dashboards

Visual progress summaries representing a larger data set.



EARNED VALUE

Measure Performance Objectively

EVERYTHING ELSE IS DERIVED FROM THESE

The Three Base Values

1

PV — Planned Value

The authorised budget assigned to the work SCHEDULED to be done by now.

2

EV — Earned Value

The budget for the work ACTUALLY COMPLETED, in budget terms.

3

AC — Actual Cost

The cost actually INCURRED for the work performed.

Variance Analysis

Cost

- Cost Variance
 - $CV = EV - AC$
 - POSITIVE is good — under budget
 - NEGATIVE is bad — over budget
 - Measured in dollars

Schedule

- Schedule Variance
 - $SV = EV - PV$
 - POSITIVE is good — ahead of schedule
 - NEGATIVE is bad — behind schedule
 - Measured in dollars, not days

Performance Indices

Cost

- Cost Performance Index
 - $CPI = EV \div AC$
 - Greater than 1 is good
 - Less than 1 means you are over budget
 - CPI of 0.84 means you get 84 cents of value per dollar spent

Schedule

- Schedule Performance Index
 - $SPI = EV \div PV$
 - Greater than 1 is good
 - Less than 1 means you are behind schedule
 - SPI of 0.80 means you are progressing at 80% of the planned rate

Forecasting

1

BAC — Budget At Completion

The total approved budget for the project.

2

EAC — Estimate At Completion

What the project will now cost in total. $EAC = BAC \div CPI$.

3

ETC — Estimate To Complete

What remains to be spent. $ETC = EAC - AC$.

4

VAC — Variance At Completion

How far off budget you will finish. $VAC = BAC - EAC$.

Earned Value Analysis with GenAI

ACTIVITY 23

Earned value is the one technique that tells you objectively whether you are ahead or behind, over or under. You will compute it by hand first, then test whether GenAI gets it right.

You'll produce

A complete earned value analysis with variances, indices and a forecast.

Tools: GenAI tool · calculator

Step-by-step instructions: [see your Learner Guide](#)

Earned Value Analysis with GenAI

ACTIVITY 23

PROMPT GenAI WITH

Using the supplied baseline, completion and actual-cost data, calculate PV, EV, AC, CV, SV, CPI, SPI, EAC and ETC. Show the formula and substitution for every result, then interpret the project status.

⚠ THEN CHECK IT

GenAI gives you a fast first draft. Check every figure, source and assumption before you use it.

Earned Value Analysis with GenAI

Check your work

You can compute PV, EV, AC, CV, SV, CPI, SPI and EAC unaided for a fresh scenario, and you found at least one error in the AI's version.



CONTROL QUALITY

Control Quality and Close the Project

Quality Control

1

Assess approaches

Evaluate the quality of project approaches and activities.

2

Inspect and test

Evaluate deliverable quality through inspection and testing.

3

Review and audit

Evaluate project activities and processes through reviews and audits.

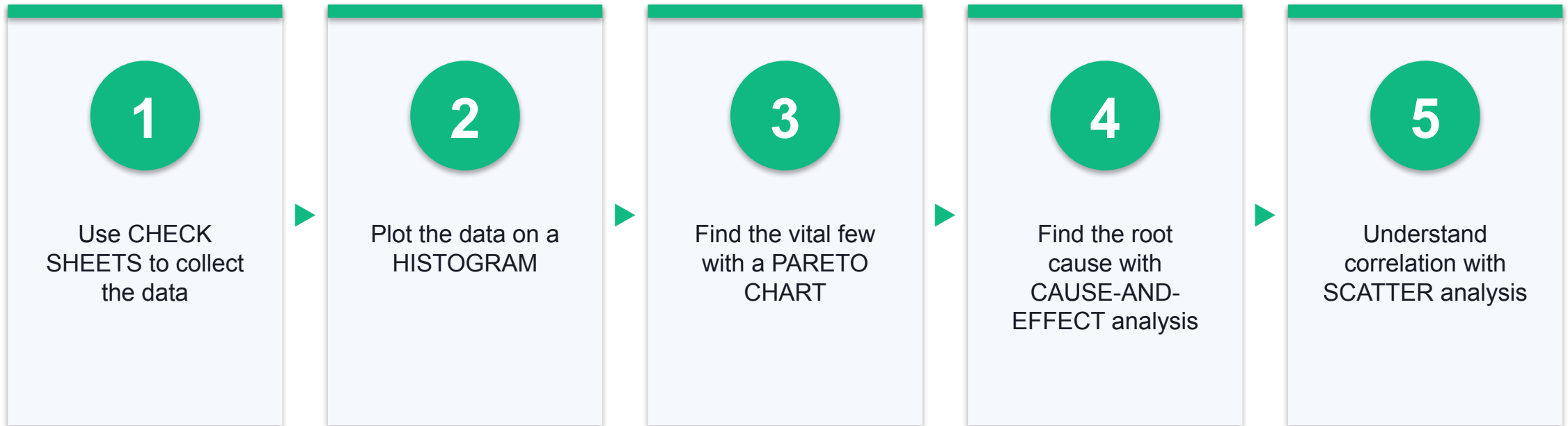
4

Detect and prevent

Focus on detecting AND preventing errors and defects.

FIVE STEPS, IN ORDER

The Control Quality Process



Control Charts

1

What it shows

The predictability, behaviour and stability of a process over time.

2

Best for

Repetitive processes with predictable results.

3

Control limits

Come from the PROCESS — what it actually does.

4

Specification limits

Come from the CUSTOMER — what they require. A process can be in control and still fail spec.

5

The rule of seven

Seven consecutive points on one side of the mean, or seven rising or falling, indicates a trend.

6

Correlation chart

Shows the relationship between two variables — a process element against a quality defect.

Control Quality with GenAI — Check Sheets and Control Charts

ACTIVITY 24

Quality control is about detecting problems from data rather than opinion. You will collect data on a check sheet and plot a control chart, applying the rule of seven.

You'll produce

A check sheet of collected data and a control chart with limits and a trend assessment.

Tools: GenAI tool · Check Sheets.xlsx · Control Chart.xlsx

Step-by-step instructions: [see your Learner Guide](#)

Control Quality with GenAI — Check Sheets and Control Charts

ACTIVITY 24

PROMPT GenAI WITH

Use the supplied defect log to produce a check-sheet summary and control-chart data. Calculate the centre line and control limits, identify any special-cause signal and explain what additional evidence is needed.

⚠ THEN CHECK IT

GenAI gives you a fast first draft. Check every figure, source and assumption before you use it.

Control Quality with GenAI — Check Sheets and Control Charts

Check your work

Your control chart has a mean and both control limits, you have checked for runs of seven, and you can distinguish control limits from specification limits.

Monitoring Risk Through Execution

1

Continuously monitor

Status, probability and impact of every open risk.

2

Identify new risks

New risks emerge as the project and its environment change.

3

Reassess and close

Reassess current risks; close risks that are now outdated.

4

Update the reserve

Track how much contingency remains against how much risk remains.

Why Projects or Phases Close

1

Successful completion

Stakeholders accept the deliverables against the acceptance criteria.

2

Requirements change

The need the project answered no longer exists.

3

No longer feasible

The deliverable cannot be produced as envisaged.

4

Business case changes

An internal decision removes the justification.

5

Legal or regulatory change

An external change prohibits progress.

6

Funding or risk

Financial support ends, or risks make completion impossible.

Close Project or Phase — Activities

1

Customer acceptance of the deliverables

2

Transition of the product to the customer

3

Notify enterprise functions; update organisational process assets

4

Prepare the final report

5

Conclude external obligations — legal, regulatory, contractual

6

Close all accounts in the financial system

7

Archive all project information

8

Release resources — people, financial and physical

Transition and Handover

1

Training

End users need training on the new product or service.

2

Documentation

The product and deliverable documentation must be handed over.

3

Communication

Effective communication between the project team and the organisation.

4

Hypercare

Post-implementation support during the period immediately after go-live.

Finalise Lessons Learned

1

Scope changes and why they happened

2

Schedule impacts and their causes

3

Risks and issues that materialised

4

Stakeholder relationships

5

Vendor relationships and performance

6

Artifacts worth reusing and recommendations

Monitor, Control Change and Close the Project with GenAI

ACTIVITY 25

The final activity runs the end of the project lifecycle: processing a change request through integrated change control, then closing the project and capturing lessons learned with GenAI's help.

You'll produce

A processed change request with impact analysis, and a project closure report.

Tools: GenAI tool · Digital AI Transformation Roadmap Studio (alfredang.github.io/digitaltransformation) · Change Request template

Step-by-step instructions: see your **Learner Guide**

Monitor, Control Change and Close the Project with GenAI

ACTIVITY 25

PROMPT GenAI WITH

Analyse each change request for its effect on scope, schedule, cost, quality, resources and risk. State confidence in each estimate, recommend an approve, reject or defer decision, and draft the closure actions for approved changes.

⚠ THEN CHECK IT

GenAI gives you a fast first draft. Check every figure, source and assumption before you use it.

Monitor, Control Change and Close the Project with GenAI

Check your work

Your change request is correctly classified with impacts on all five dimensions, it has been through the CCB path, and your closure report covers acceptance, transition, contracts, archiving and resource release.

Recap — Managing Project Risks with GenAI

- Anticipate potential risks to project delivery (A3).
- Prioritise risks by probability and impact for further action.
- Quantify risk exposure to support a financial decision.
- Devise risk mitigation measures to ensure project success (A4).
- Select the right contract type and manage procurement risk.
- Measure project performance objectively and forecast the outcome.
- Detect whether a process is stable and in control.



WRAP-UP

Course Summary & Next Steps

What You Achieved

1

LO1 · Initiation & Planning

Charters, business cases, stakeholder maps, RACI, requirements, WBS, schedules and budgets — all built with GenAI.

2

LO2 · Execution & Delivery

Performance reporting, knowledge capture, root cause analysis, resource and budget optimisation, team and communications.

3

LO3 · Risk & Closure

Risk identification and analysis, response planning, procurement, change control, earned value and project closure.

4

The habit that matters

You prompt for a first draft, then verify every figure, source and assumption before it leaves your hands.

NEXT STEPS

Applying This at Work Next Week

1

Start small

Pick one artifact — a status report or a risk register — and draft it with GenAI.

2

Reuse your prompt library

The prompts you built in Activity 1 work on your real projects.

3

Check your policy

Confirm what your organisation allows you to put into a GenAI tool.

4

Keep the ed-tools

RACI, BMC, Design Thinking, Scrum and the Roadmap Studio are free and need no install.

5

Verify everything

Especially arithmetic, sources and anything about a named person.

6

Share the practice

Show one colleague the prompting pattern that helped you most.

Course Feedback

1

Why it matters

Your feedback tells us what worked and what to improve for the next run.

2

Where

Complete the feedback form on the LMS at lms-tms.tertiaryinfotech.com

3

Your certificate

Your certificate is issued through the same portal.

4

When

Please complete it before you leave the classroom.

Recommended Next Courses

1

Agile Project Management for Business

2

Mastering Agile Project Management for IT Projects

3

Business Analysis with Generative AI

4

Data Analytics with Generative AI

5

Generative AI for Productivity

WE'RE HERE TO HELP

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SEE YOU AT THE NEXT COURSE

Thank You!

You can now plan, execute and control a project with GenAI as your co-pilot — and you know exactly where its judgement ends and yours begins.